

**“DenguScope: Identification of Transcriptomic Biomarkers
for Dengue Infection”**

by

Ms. S Varsha

Reg. No. 24L10930

Dissertation submitted to



JSS Academy of Higher Education & Research, Mysuru

In partial fulfillment of the requirements

for the Degree

of

MASTER OF SCIENCE

in

BIOINFORMATICS

Under the guidance of

DR. KIRAN KUMAR M N

Associate Professor

Department of Biotechnology and Bioinformatics

School of Life Sciences

JSS Academy of Higher Education & Research

Mysuru - 570015

JUNE 2026

**“DenguScope: Identification of Transcriptomic Biomarkers
for Dengue Infection”**

by

Ms. S Varsha

Reg. No. 24L10930

Dissertation submitted to



JSS Academy of Higher Education & Research, Mysuru

In partial fulfillment of the requirements

for the Degree

of

MASTER OF SCIENCE

in

BIOINFORMATICS

Under the guidance of

DR. KIRAN KUMAR M N

Associate Professor

Department of Biotechnology and Bioinformatics

School of Life Sciences

JSS Academy of Higher Education & Research

Mysuru - 570015

JUNE 2026



Department of Biotechnology & Bioinformatics
School of Life Sciences
JSS Academy of Higher Education & Research
Mysuru, Karnataka, India - 570015
(Deemed to be University)
Accredited "A⁺⁺" Grade by NAAC

SUBMISSION / EVALUATION CERTIFICATE

This is to certify that this dissertation entitled “**DenguScope: Identification of Transcriptomic Biomarkers for Dengue Infection**” has been submitted by **Ms. S Varsha** to the JSS Academy of Higher Education & Research, Mysuru under the guidance of **DR. KIRAN KUMAR M N**, Associate Professor, Department of Biotechnology and Bioinformatics, School of Life Sciences, JSS Academy of Higher Education & Research, Mysuru in partial fulfillment of the requirements for the degree of **Master of Science in Bioinformatics**.

Date:

Controller of Examinations

This is to certify that this dissertation has been evaluated.

Signature of Examiners	1. 2.
Names:	1. 2.
Date	



Department of Biotechnology & Bioinformatics
School of Life Sciences
JSS Academy of Higher Education & Research
Mysuru, Karnataka, India - 570015
(Deemed to be University)
Accredited "A⁺⁺" Grade by NAAC

DECLARATION BY THE CANDIDATE

I, **Ms. S Varsha** hereby declare that this dissertation entitled **“DenguScope: Identification of Transcriptomic Biomarkers for Dengue Infection”** submitted in partial fulfillment of the requirements for the award of the degree of **Master of Science** in **Bioinformatics**, is a record of original research work carried out by me under the guidance of **DR. KIRAN KUMAR M N**, Associate Professor, Department of Biotechnology and Bioinformatics, School of Life Sciences, JSS Academy of Higher Education & Research, Mysuru, Karnataka.

This dissertation has not formed the basis for the award of any other Degree/ Diploma/ Associateship/ Fellowship or another similar title of any candidate of any University.

Date:
Place: Mysuru

S Varsha
Postgraduate Student
Department of Biotechnology and Bioinformatics
School of Life Sciences
JSS Academy of Higher Education & Research
Mysuru - 570015



Department of Biotechnology & Bioinformatics
School of Life Sciences
JSS Academy of Higher Education & Research
Mysuru, Karnataka, India - 570015
(Deemed to be University)
Accredited "A++" Grade by NAAC

CERTIFICATE BY THE GUIDE

This is to certify that this dissertation entitled **“DenguScope: Identification of Transcriptomic Biomarkers for Dengue Infection”** is a bonafide research work done by **Ms. S Varsha**, Postgraduate student, Department of Biotechnology and Bioinformatics, School of Life Sciences, JSS Academy of Higher Education & Research, Mysuru, under my guidance in partial fulfillment of the requirements for the degree of **Master of Science in Bioinformatics**.

Date:
Place: Mysuru

DR. KIRAN KUMAR M N
Associate Professor & Guide
Department of Biotechnology and Bioinformatics
School of Life Sciences
JSS Academy of Higher Education & Research
Mysuru - 570015



Department of Biotechnology & Bioinformatics
School of Life Sciences
JSS Academy of Higher Education & Research
Mysuru, Karnataka, India - 570015

(Deemed to be University)

Accredited "A⁺⁺" Grade by NAAC

ENDORSEMENT BY THE COORDINATOR OF THE DEPARTMENT

This is to certify that this dissertation entitled **“DenguScope: Identification of Transcriptomic Biomarkers for Dengue Infection”** is a bonafide research work done by **Ms. S Varsha**, Postgraduate student, Department of Biotechnology and Bioinformatics, School of Life Sciences, JSS Academy of Higher Education & Research, Mysuru, under the guidance of **DR. KIRAN KUMAR M N**, Associate Professor, Faculty of Life Sciences, Department of Biotechnology and Bioinformatics, School of Life Sciences, JSS Academy of Higher Education & Research, Mysuru and **DR. HIMANSHU KUMAR**, Professor, Laboratory of Immunology and Infectious Diseases Biology, Department of Biological Sciences, Indian Institute of Science Education and Research (IISER) Bhopal in partial fulfillment of the requirements for the Degree of **Master of Science in Bioinformatics**.

COORDINATOR

Department of Biotechnology and Bioinformatics
School of Life Sciences
JSS Academy of Higher Education & Research
Mysuru - 570015

ENDORSEMENT BY THE HEAD OF THE DEPARTMENT

This is to certify that this dissertation entitled **“DenguScope: Identification of Transcriptomic Biomarkers for Dengue Infection”** is a bonafide research work done by **Ms. S Varsha**, Postgraduate student, Department of Biotechnology and Bioinformatics, School of Life Sciences, JSS Academy of Higher Education & Research, Mysuru, under the guidance of **DR. KIRAN KUMAR M N**, Associate Professor, Department of Biotechnology and Bioinformatics, School of Life Sciences, JSS Academy of Higher Education & Research, Mysuru and **DR. HIMANSHU KUMAR**, Professor, Laboratory of Immunology and Infectious Diseases Biology, Department of Biological Sciences, Indian Institute of Science Education and Research (IISER) Bhopal in partial fulfillment of the requirements for the Degree of **Master of Science in Bioinformatics**.

PROFESSOR & HEAD

School of Life Sciences
JSS Academy of Higher Education & Research
Mysuru - 570015

CERTIFICATE OF ANTI - PLAGIARISM CHECK

It is certified that this dissertation titled **“DenguScope: Identification of Transcriptomic Biomarkers for Dengue Infection”** by **Ms. S Varsha**, Postgraduate student, Department of Biotechnology and Bioinformatics, School of Life Sciences, JSS Academy of Higher Education & Research, Mysuru, has been subjected to anti-plagiarism check.

We undertake the following:

- The research work submitted is original.
- No sentence, table, paragraph or section has been copied verbatim from any other resource unless it is placed under quotation marks and duly referenced.
- There is no fabrication of data or results which have been compiled.
- There is no falsification by manipulating research materials, equipment or processes.
- The dissertation has been checked using Anti-plagiarism web tool “TURNITIN” and found within limits as per the anti-plagiarism guidelines of JSS AHER, Mysuru.
- Similarity index of the dissertation is as follows:

% Similarity index	10 %
Total continuous words	4831

	Name	Signature
Guide:	Dr. Kiran Kumar M N	
Student:	S Varsha	

Recommendation/ comments by Department Academic Integrity Panel (DAIP):

	DAIP	Name	Signature
1.	Chairman		
2.	Member		
3.	Member Secretary		

Institution seal:

Acknowledgement

*I would like to express my sincere gratitude to **His Holiness Jagadguru Sri Shivarathri Deshikendra Mahaswamiji**, Chancellor, JSS Academy of Higher Education & Research, for providing an excellent academic environment and encouragement towards research and education.*

*I extend my respectful thanks to **Dr. B. Suresh**, Pro Chancellor, **Dr. H. Basavanagowdappa**, Vice Chancellor, and **Dr. Manjunatha B.**, Registrar, JSS Academy of Higher Education & Research.*

*I would also like to acknowledge **Dr. Sudeendra Bhat R.**, Controller of Examinations, **Dr. Vishal Kumar Gupta**, Dean (Academics), and **Dr. Prashant M. Vishwanath**, Dean (Research), JSS Academy of Higher Education & Research.*

*I express my heartfelt gratitude to **Dr. Raveesha K A**, Dean (School of Life Sciences), JSS Academy of Higher Education and Research, Mysuru, for his constant encouragement and support.*

*I am deeply grateful to **Dr. Gopenath T S**, Professor and Coordinator, Department of Biotechnology and Bioinformatics, School of Life Sciences, JSS Academy of Higher Education and Research, Mysuru.*

*I sincerely thank **Dr. Ramith Ramu**, PG Coordinator, Department of Biotechnology and Bioinformatics, School of Life Sciences, JSS Academy of Higher Education and Research, Mysuru.*

*I would like to express my profound gratitude to **Dr. Kiran Kumar M N**, Associate Professor and Internal Guide, Department of Biotechnology and Bioinformatics, JSS Academy of Higher Education and Research, Mysuru for his invaluable guidance, continuous encouragement and support throughout the dissertation.*

*I would like to express my sincere gratitude to **Dr. Himanshu Kumar**, Professor and External Guide, Laboratory of Immunology and Infectious Diseases Biology, Department of Biological Sciences, Indian Institute of Science Education and Research (IISER) Bhopal, for providing me with the opportunity to work in his laboratory. I am deeply thankful for his constant encouragement, support, and guidance throughout my dissertation work.*

*I would also like to extend my heartfelt thanks to **Dr. Pandikannan Krishnamoorthy**, Project Research Scientist, Prof. Himanshu Kumar's Group, Laboratory of Immunology and Infectious Diseases Biology, Indian Institute of Science Education and Research (IISER) Bhopal, for his invaluable guidance throughout my dissertation. The knowledge, expertise, and research experience he shared greatly enriched my learning during this short period.*

*I am grateful to all the **members of the Laboratory of Immunology and Infectious Diseases Biology**, Indian Institute of Science Education and Research (IISER) Bhopal, for their insights, support, and helpful discussions. Their guidance helped me gain a deeper understanding of the research process and provided me with valuable exposure to a professional research environment.*

*I want to express my gratitude to **everyone at the Department of Biotechnology and Bioinformatics**, for their support throughout the course of my dissertation. I express my thanks to everyone at JSS Academy of Higher Education and Research for their guidance, support and cooperation in every aspect.*

*Lastly, I would like to thank my **family members and friends** for their unwavering support and encouragement throughout this dissertation.*

LIST OF ABBREVIATIONS

AST - Aminotransferase (Liver Enzyme Marker)
DESeq2 - Differential Expression Analysis Tool (R package)
DNA - Deoxyribonucleic Acid
ELISA - Enzyme-Linked Immunosorbent Assay
EMBL-EBI - European Molecular Biology Laboratory - European Bioinformatics Institute
FDR - False Discovery Rate
GC - GC Content (Guanine-Cytosine Content)
GEO - Gene Expression Omnibus
GSE - GEO Series Accession
HISAT - Hierarchical Indexing for Spliced Alignment of Transcripts
HTML - HyperText Markup Language
HTSeq - High-throughput Sequencing Tool for Read Counting
IgG - Immunoglobulin G
IgM - Immunoglobulin M
IL-6 ⁺ - Interleukin-6 Positive Cells
IFN- γ ⁺ CD56 ⁺ CD3 ⁺ NKT cells - Interferon Gamma Positive Natural Killer T Cells
LC-MS - Liquid Chromatography-Mass Spectrometry
MMSEQ - Mixture Model for Expression Quantification
NS1 - Non-Structural Protein 1 (Antigen)
PBMC - Peripheral Blood Mononuclear Cells
RNA - Ribonucleic Acid
RNA-seq - Ribonucleic Acid Sequencing
RPKM/FPKM - Reads/Fragments Per Kilobase of transcript per Million mapped reads
RSEM - RNA-Seq by Expectation Maximization
RT-PCR - Reverse Transcription Polymerase Chain Reaction
TMM - Trimmed Mean of M-values (Normalization Method)
TV003 - Dengue Vaccine Candidate (Tetravalent Vaccine Strain)

ABSTRACT

Dengue fever is a major global public health concern, with clinical outcomes ranging from mild febrile illness to severe and potentially life-threatening disease. Early and accurate identification of disease severity is essential for effective patient management, but current diagnostic methods such as NS1 antigen detection, RT-PCR, and serological assays are limited by factors including infection stage, turnaround time, and sensitivity. In this study, transcriptomic data were analyzed to investigate host gene expression changes associated with dengue infection and disease severity.

RNA sequencing data were processed to identify differentially expressed genes across different clinical conditions, followed by statistical filtering to ensure the reliability of the results. Functional enrichment analyses, including Gene Ontology and pathway analysis, were then performed to understand the biological roles of these genes. The results showed clear differences in gene expression patterns between disease states. Genes involved in immune response, inflammatory signaling, cytokine activity, and cellular stress pathways were found to be significantly dysregulated, indicating their role in dengue pathogenesis. These pathways are known to be closely associated with host - virus interactions and disease progression.

Further analysis identified a subset of consistently altered genes that showed strong potential as candidate biomarkers for distinguishing between mild and severe dengue cases. These genes may help improve the accuracy of disease stratification and support earlier clinical decision-making. Overall, this study demonstrates that transcriptomic profiling is a useful approach for capturing host response signatures in dengue infection. The identified gene expression patterns provide promising biomarker candidates that could contribute to the development of improved diagnostic and prognostic tools, ultimately aiding better management of dengue patients.

Key Words:

Dengue fever; transcriptomic analysis; RNA sequencing; differentially expressed genes; host gene expression; biomarkers; disease severity; immune response; cytokine signaling; gene expression profiling; functional enrichment analysis; Gene Ontology; pathway analysis.

TABLE OF CONTENTS

Chapter No.	Title	Page No.
I	INTRODUCTION	1
II	REVIEW OF LITERATURE	10
III	BIONFORMATICS DATABASES AND TOOLS USED	14
IV	METHODOLOGY	19
V	RESULTS	24
VI	CONCLUSION	36
	REFERENCES	37

LIST OF FIGURES

Figure No.	Title	Page No.
1	Gene Expression changes lead to changes in Cell Activity	1
2	Structure and serotype of the dengue virus	2
3	Dengue virus replication cycle	3
4	Dengue clinical progression	4
5	Types of Coding & Non-Coding RNAs	6
6	Library preparation methods for different RNA-seq methods	7
7	RNA-sequencing technologies	7
8	RNA-seq data analysis workflow for differential gene expression	8
9	Sequential searching of only unmapped portions of reads	14
10	Clustering of seeds that are not multi-mapping	14
11	DESeq2 Steps	15
12	Dataset Collection steps	17
13	Quality Control steps	18
14	Genome Indexing steps	19
15	Alignment steps	20
16	Quantification steps	20
17	Differential Expression Analysis steps	21
18	Gene Ontology (GO) Enrichment steps	22
19	KEGG Pathway Analysis steps	23
20	R Shiny App Development	23
21	Sample Distribution based on gender, age, platelet count ($10^3/\mu\text{L}$), and sample type	24
22	Principal Component Analysis (PCA) plot	25

23	MA Plot of Differential Gene Expression	26
24	Gene Expression of the Top Upregulated Gene IGHG1 across groups	27
25	Gene Expression of the Top Downregulated Gene RN7SKP9 across groups	27
26	Gene Expression of the Top Upregulated Gene IGHG1 across samples	28
27	Gene Expression of the Top Downregulated Gene RN7SKP9 across samples	28
28	Distribution of Differentially Expressed Genes	29
29	Volcano Plot of Differentially Expressed Genes	31
30	Heatmap of Top 20 Differentially Expressed Genes	32
31	Dot Plot of Immune related Gene Ontology Terms	33
32	Dot Plot of Immune related KEGG Pathways	34
33	Shiny App Homepage	34

LIST OF TABLES

Table No.	Title	Page No.
1	Top 20 Significant Upregulated Genes	30
2	Top 20 Significant Downregulated Genes	31

CHAPTER I

INTRODUCTION

1. Overview

Dengue, a mosquito-borne viral infection, affects millions of people worldwide, but why do individuals respond to it differently? Even when people are infected under similar environmental conditions, why do some develop mild symptoms while others experience severe disease? This variation is influenced by a combination of internal factors such as genetics, immune strength, age, and prior dengue infection, as well as external factors like viral load, virus serotype, and environmental conditions. Due to these differences, a single standard treatment approach is often not sufficient for all patients. Current clinical management mainly focuses on controlling symptoms, but there is a growing need to understand the disease at a deeper, molecular level.

But then another question arises: if every cell in our body contains the same DNA, why do different cells behave so differently? How does a liver cell differ from a brain or muscle cell if their genetic material is identical? The answer lies in gene expression. Different cells switch on different sets of genes depending on their function, tissue type, and stage of development. This makes gene expression both spatial (varying across tissues) and temporal (changing over time). Only the genes needed by a particular cell at a particular time are expressed. DNA is transcribed into RNA, and RNA is translated into proteins, which carry out most cellular functions. Therefore, while DNA represents all the genes that an organism possesses, RNA and proteins represent the genes that are actually being expressed.

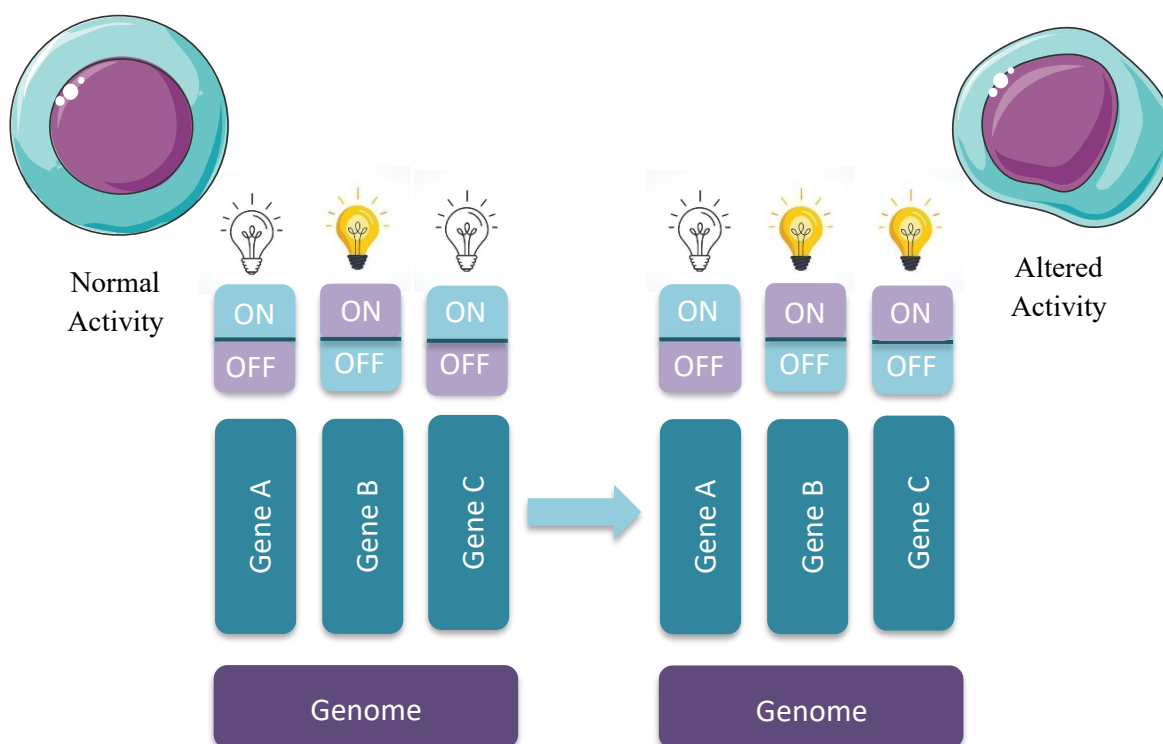


Figure 1: Gene Expression changes lead to changes in Cell Activity

This is where RNA sequencing (RNA-seq) is useful. RNA-seq allows us to identify which genes are being expressed in a sample and at what levels. By comparing healthy individuals with dengue patients, we can identify differentially expressed genes that help us understand how dengue affects the body and how the immune system responds. They can also point to useful biomarkers and possible treatment targets, which may help predict disease severity and support better personalized treatments.

2. Dengue

2.1. Dengue Virus and its structure

Dengue is a viral infection caused by the dengue virus (DENV), which belongs to the genus *Flavivirus* in the family *Flaviviridae*. There are four different types of the virus called DENV-1, DENV-2, DENV-3, and DENV-4. These are called serotypes because each one reacts differently with the body's immune system, even though they are genetically quite similar and have different versions that can affect disease severity.

The dengue virus is a small, spherical virus that contains a single-stranded RNA genome about 11,000 bases long. This RNA produces viral proteins, including structural proteins (capsid, membrane, and envelope) that form the virus and non-structural proteins that help it replicate. The virus is surrounded by a lipid envelope containing envelope and membrane proteins, which enable it to attach to and enter host cells.

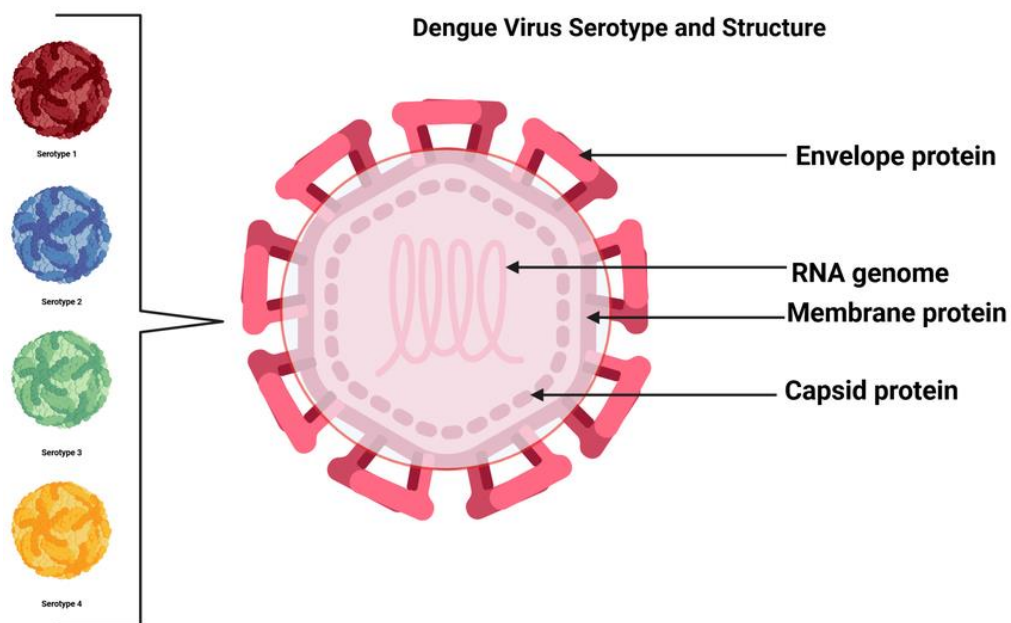


Figure 2: Structure and serotype of the dengue virus [1]

2.2. Dengue Virus Transmission and Life Cycle

Dengue virus is mainly transmitted to humans through the bite of an infected female *Aedes aegypti* mosquito. After feeding on an infected person, the virus multiplies inside the mosquito and eventually reaches its salivary glands, making the mosquito capable of transmitting the

virus to another person. Once an infected mosquito bites a human, the virus enters the body and infects immune cells by attaching to receptors on the cell surface through its envelope (E) protein.

The virus enters the host cell through endocytosis and releases its RNA into the cytoplasm. The viral RNA then uses the host cell machinery to produce viral proteins and replicate its genome. Newly formed viral proteins and RNA assemble into immature virus particles in the endoplasmic reticulum. These particles are transported to the Golgi apparatus, where they mature before being released from the cell through exocytosis. The released viruses can then infect other cells, allowing the infection to spread throughout the body.

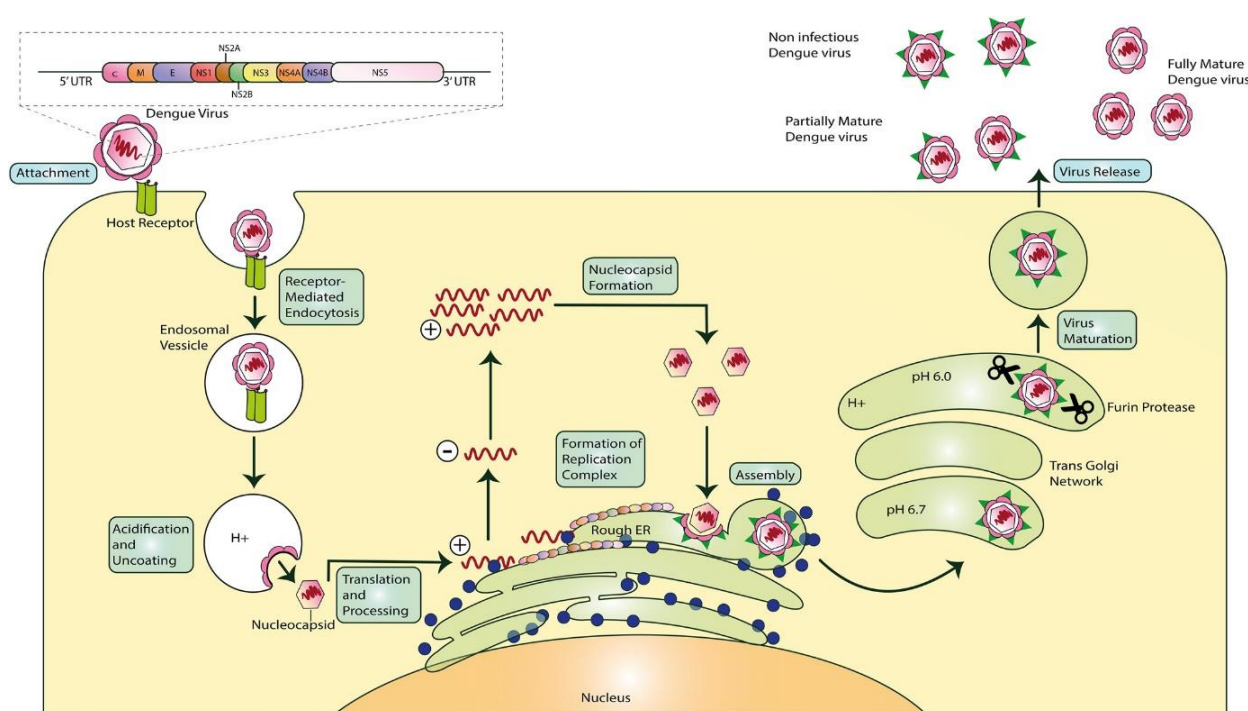


Figure 3: Dengue virus replication cycle [2]

2.3. Clinical Progression of Dengue Infection

The World Health Organization (WHO) classifies dengue into three categories: dengue without warning signs, dengue with warning signs, and severe dengue. Most infections are mild or asymptomatic, with about 60–80% of infected individuals showing no symptoms or only mild illness. In symptomatic cases, the disease typically progresses through three stages: the febrile phase, the critical phase, and the recovery phase.

2.3.1 Febrile Phase

The febrile phase is the first stage of dengue and usually lasts 2–7 days. It starts with a sudden high fever and is often accompanied by headache, muscle and joint pain, nausea, vomiting, rash, and tiredness.

2.3.2 Critical Phase

Some dengue patients enter the critical phase, usually around days 4–6, when the fever starts to decrease. During this phase, fluid can leak from blood vessels into surrounding tissues, called plasma leakage. This can lead to fluid accumulation, shock, and sometimes bleeding, and is the main cause of severe dengue. Warning signs include severe stomach pain, repeated vomiting, bleeding from gums or nose, unusual tiredness or restlessness, and low platelet count.

A small percentage of patients develop severe dengue, which can be life-threatening. It can cause severe bleeding, shock due to poor blood circulation, or damage to organs like the liver, heart, or brain.

2.3.3 Recovery Phase

During the recovery phase, fluid that leaked into tissues is reabsorbed and the patient gradually improves. Some patients may develop a characteristic rash with red areas and “white islands” of normal skin, and adults may experience fatigue or mood changes for weeks after recovery.

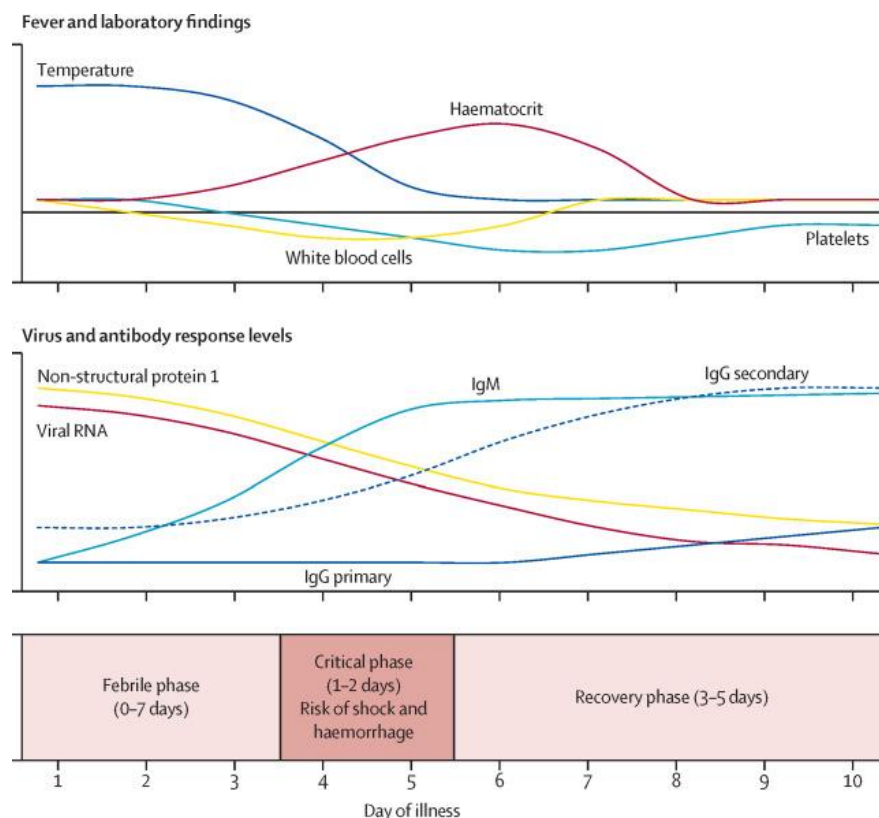


Figure 4: Dengue clinical progression [3]

2.4. Diagnosis of Dengue

Diagnosis of dengue depends on the stage of infection. In the early phase (0–7 days), tests like nucleic acid amplification tests (NAAT) are used, which are the most accurate because they detect viral RNA and can also identify the virus type. The NS1 antigen test is also helpful during this time, although it may be less sensitive in secondary infections.

From about day 4 onwards, blood tests that detect antibodies are more useful. IgM antibodies show a recent infection and can stay in the blood for up to 12 weeks, while IgG antibodies indicate past infection and may remain for life. In secondary infections, IgG levels rise quickly in the early stage. However, these antibody tests can sometimes react with similar viruses, which may affect accuracy.

2.5. Management of Dengue

There is currently no specific antiviral treatment for dengue, so management is mainly supportive and follows WHO guidelines based on disease severity. Mild cases are treated at home with oral fluids, paracetamol for fever, and close monitoring for warning signs. Patients with warning signs or other health problems may need hospital care and intravenous fluids. Severe dengue requires treatment in intensive care, including careful fluid replacement, possible blood transfusion, and close monitoring of vital signs and organ function. Fluid management is a critical aspect of treatment as both underhydration and overhydration can lead to complications.

2.6. Dengue Vaccines and Prevention (Vector Control)

There are vaccines for dengue, such as Dengvaxia, Qdenga, and TV003, but they are not widely used as they do not protect equally against all dengue types, may have safety concerns, and some require confirmation of past infection before use.

Prevention mainly focuses on controlling mosquitoes. This includes using mosquito repellents, wearing full-sleeve clothing, using bed nets, and removing standing water where mosquitoes breed. New approaches like Wolbachia-infected mosquitoes and genetically modified mosquitoes are also being explored to help reduce dengue spread.

3. RNA

3.1 Types of RNA

Cells contain different types of RNA, each with a specific role. mRNA (messenger RNA) carries genetic information from DNA to make proteins. rRNA (ribosomal RNA) forms the main structure of ribosomes, while tRNA (transfer RNA) brings amino acids during protein synthesis. miRNA (microRNA) helps control gene expression, and snRNA (small nuclear RNA) is involved in RNA splicing. circRNA (circular RNA) forms a circular structure and is more stable, making it an important area of research.

Other important regulatory RNAs include lncRNA (long non-coding RNA), which regulates gene expression at different levels such as chromatin modification, transcription, and post-transcriptional control. snoRNA (small nucleolar RNA) helps modify and process rRNA inside the nucleolus. smRNA (small RNA) is a general term for short regulatory RNAs involved in gene silencing and gene regulation processes.

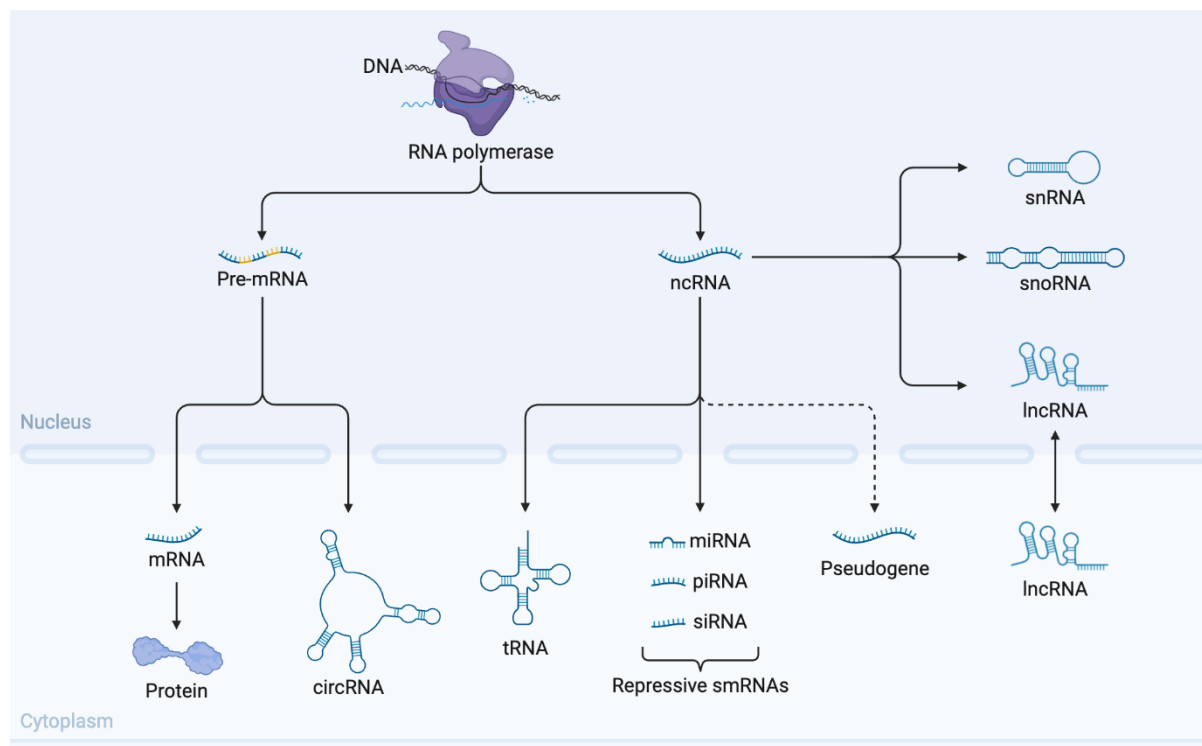


Figure 5: Types of Coding & Non-Coding RNAs [from biorender.com]

3.2 RNA Sequencing Workflow

RNA sequencing begins by extracting total RNA from cells or tissues. Since most of this RNA is ribosomal RNA (over 80%), it is usually removed, and mRNA is enriched using poly(A) selection or rRNA depletion methods. The RNA is then converted into complementary DNA (cDNA) using reverse transcription as DNA is more stable. The cDNA is fragmented, and short adapter sequences are added to both ends to prepare a sequencing library. The library is then sequenced to generate millions of short reads, which are then aligned to a reference genome or transcriptome to identify the genes they originate from.

3.3 RNA Sequencing Technologies and strategies

RNA-seq mainly uses three approaches. Short-read sequencing produces small, accurate, and low-cost reads that are widely used for gene expression studies. Long-read sequencing generates much longer reads that can cover full transcripts, making it useful for studying

isoforms and alternative splicing. Direct RNA sequencing reads RNA directly without converting it into cDNA, preserving RNA modifications and poly(A) tail information.

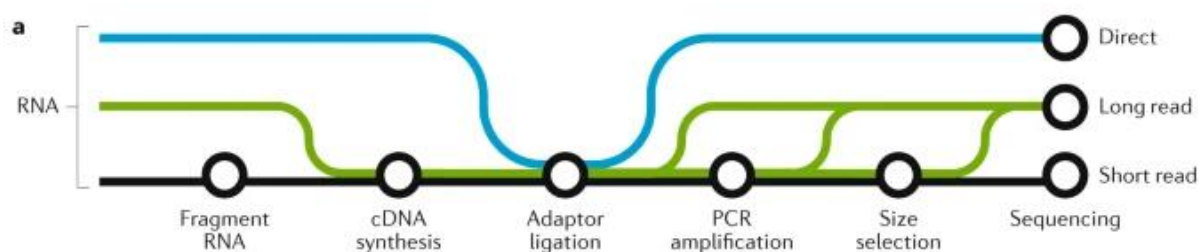


Figure 6: Library preparation methods for different RNA-seq methods [4]

Different sequencing platforms use different technologies. Illumina sequencing produces short reads (50–500 bp) using sequencing by synthesis with fluorescent signals. Pacific Biosciences (PacBio) generates very long reads (up to ~50 kb) by detecting fluorescence during nucleotide incorporation. Oxford Nanopore sequencing reads RNA or DNA molecules by measuring changes in electrical current as the strand passes through nanopores, producing reads of about 1–10 kb.

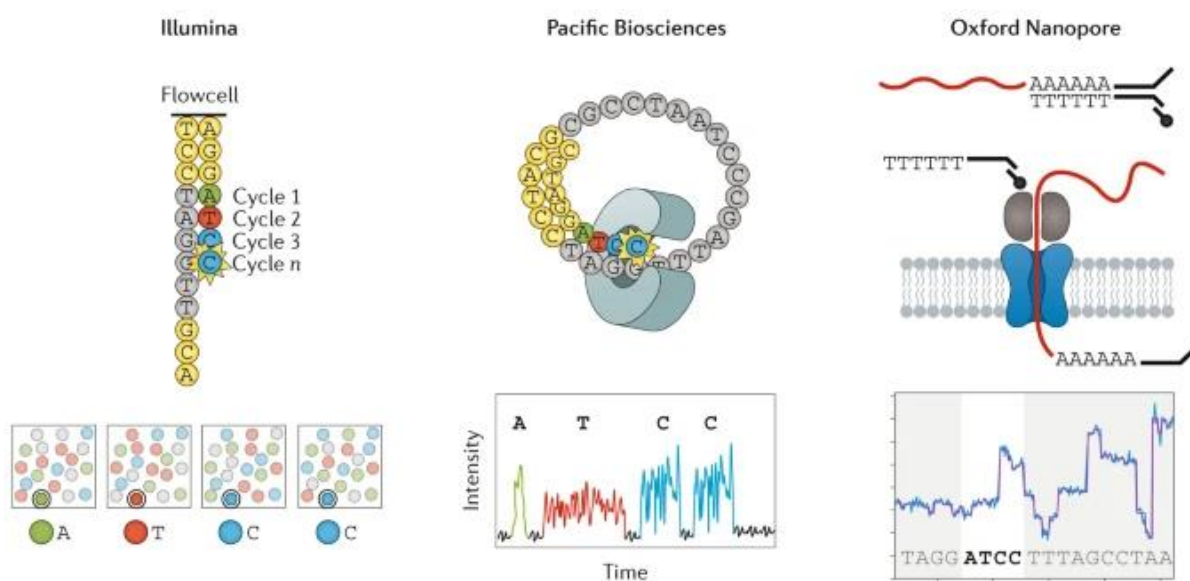


Figure 7: RNA-sequencing technologies [4]

3.4 RNA-seq Data Analysis

RNA-seq data consists of millions of raw reads that require processing before they can be interpreted biologically. The workflow varies depending on study goals and tools used, which can affect the final results.

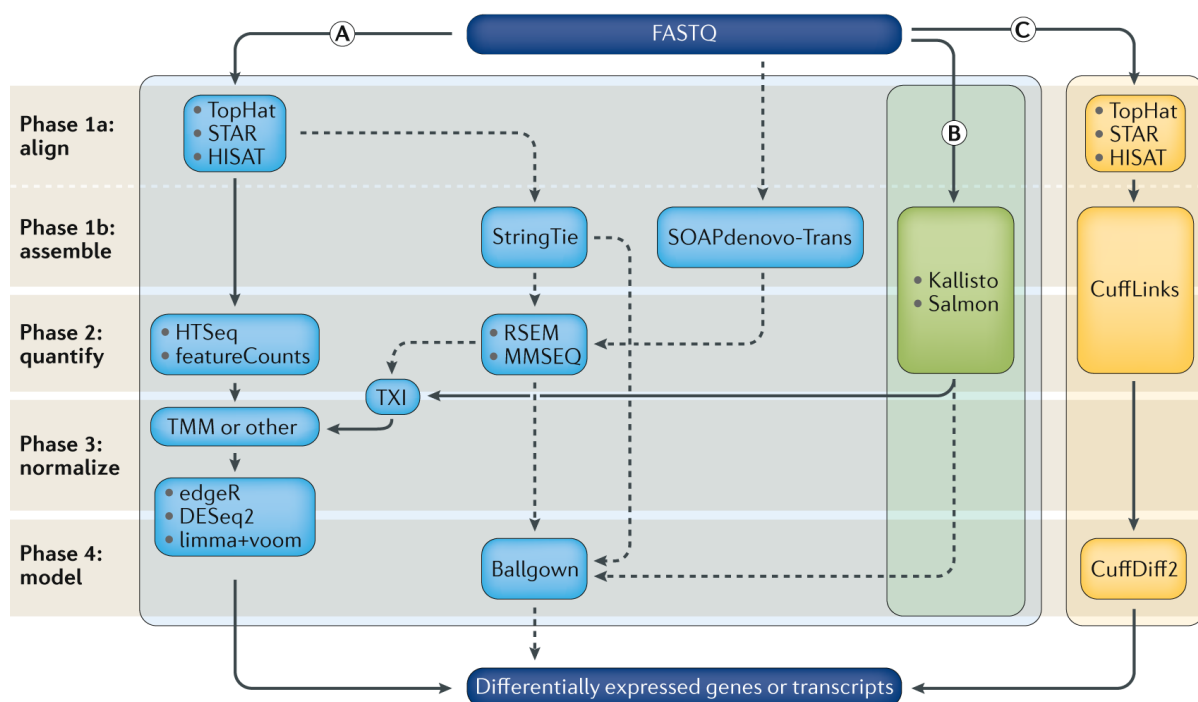


Figure 8: RNA-seq data analysis workflow for differential gene expression [4]

Phase 1: Alignment and Assembly of Sequencing Reads

Sequencing data is usually stored in FASTQ files, which contain the nucleotide sequence along with a quality score for each base. Each read has four lines: an ID, the sequence, a separator, and the quality score. Tools like STAR, HISAT2, and TopHat are commonly used for this.

First, the reference genome taken from a public database is indexed, and then the reads from the FASTQ file are aligned to it. Then the alignment rate is checked to see how well the reads matched the reference. If a high-quality reference genome is not available, a closely related species can be used, but this may reduce mapping accuracy.

In such cases, transcript assembly tools like StringTie or SOAPdenovo-Trans can help reconstruct transcripts from the aligned reads. Newer tools like Salmon, Kallisto, and Sailfish can estimate gene expression without full alignment, but they may be less accurate for low-abundance genes or short transcripts.

Phase 2: Quantification of Transcript Abundance

The next step is quantification, where gene expression is measured by counting how many reads map to each gene or transcript using a GTF annotation file which is downloaded from a public database. It is important that the reference genome and GTF file match, as using different versions can lead to incorrect gene annotations and read assignments. Gene-level analysis combines all transcript variants of a gene into one expression value, while transcript-level analysis measures each isoform separately. This is important as a single gene can produce

multiple transcripts with different functions, and some biological changes may be missed if only gene-level analysis is performed.

A challenge in this step is that many reads do not map uniquely, especially in shared or similar genomic regions. Different tools handle this in different ways, either by counting only uniquely mapped reads or by using statistical models to estimate the most likely origin of each read. Common tools for quantification include HTSeq, featureCounts, RSEM, StringTie, Kallisto, and Salmon. Some tools rely on alignments, while others estimate transcript abundance directly without full alignment. The final output is an expression matrix, where rows represent genes or transcripts and columns represent samples which is used for further downstream analyses.

Phase 3: Filtering and Normalization

Quantified gene or transcript counts are filtered and normalized to reduce technical noise and differences like sequencing depth and GC content. Filtering removes genes with very low counts, which helps improve the detection of true differential expression. Normalization is more complex. Older methods like RPKM/FPKM are now not preferred for differential expression, and have been replaced by better methods like quartile, median, and TMM (edgeR).

These methods assume that most genes are not changing between samples and that overall expression levels are similar. But sometimes these assumptions are not true, especially when many genes change in one condition. In such cases, simple read-depth correction can give wrong results. Methods like TMM help correct these differences more accurately.

Phase 4: Differential Expression modelling

After the expression matrix is generated, statistical models are used to identify genes or transcripts that are differentially expressed between conditions. Gene-level methods such as edgeR, DESeq2, and limma-voom use generalized linear models and are widely used due to their efficiency and consistent results. Transcript-level methods such as CuffDiff, MMSEQ, and Ballgown analyze isoform-level expression but are more computationally intensive and may produce more variable results.

Overall, earlier steps like alignment, quantification, filtering, and normalization have a greater impact on the final results than the choice of differential expression tool.

CHAPTER II

REVIEW OF LITERATURE

1. Literature Survey

Falconi-Agapito et al. (2022) evaluated 20 synthetic dengue virus-derived peptides to identify biomarkers for dengue diagnosis. A bead-based multiplex peptide immunoassay combined with a random forest machine learning model was used to analyze serum samples from dengue patients and individuals infected with other viruses. The study found that combining multiple peptides improved diagnostic accuracy compared to individual peptide markers. A limitation of the study is that the identified peptide biomarkers require validation in larger and more diverse populations. Since they are based on peptide responses, they may not fully reflect the early molecular changes that occur during dengue infection. [5]

Thach et al. (2021) performed a systematic review and meta-analysis of 40 observational studies involving 6,160 laboratory-confirmed dengue patients to identify early biomarkers associated with severe dengue. The study found that low platelet count and high AST levels were the most reliable biomarkers. One limitation is that both platelet count and AST are non-specific biomarkers and can be altered in many other infections and inflammatory conditions. The results also showed considerable variation across the included studies. [6]

Vasey et al. (2020) conducted a prospective cohort study involving 204 patients with suspected dengue to identify biomarkers associated with disease progression. Daily blood samples were analyzed for hematological, coagulation, biochemical, and inflammatory markers, and albumin, fibrinogen, platelet count, D-dimer, thrombin time, and AST were identified as potential biomarkers. The study is limited because these biomarkers change as the disease progresses and require repeated sample collection. This makes it difficult to identify biomarkers that remain consistent throughout the course of infection. [7]

Pradeep et al. (2021) investigated the early innate immune response in 596 laboratory-confirmed dengue patients using flow cytometry-based intracellular cytokine staining. The study identified IFN- γ^+ CD56 $^+$ CD3 $^+$ NKT cells and IL-6 $^+$ granulocytes as potential biomarkers. Although these biomarkers are promising, they represent only selected immune cell populations and do not provide a complete picture of the host immune response during dengue infection. Further validation is also needed. [8]

Moallemi et al. (2025) conducted a prospective nested case-control study using plasma samples collected within the first 96 hours of fever from 228 clinically suspected dengue patients. A 34-plex Luminex assay and ELISA identified biomarkers including VCAM-1, IL-33R, CCL11, CXCL11, and SAA that were associated with plasma leakage and disease severity. An important limitation is that the study examined only a predefined panel of

inflammatory proteins, so other important biomarkers involved in dengue infection may have been missed. The identified biomarkers also need further validation in larger populations. [9]

Shukla et al. (2023) conducted a systematic review and meta-analysis to evaluate serum ferritin as a biomarker for dengue severity. The study found that ferritin levels were significantly higher in patients with severe dengue than in those with non-severe disease. A major limitation is that ferritin is a non-specific inflammatory marker and can be elevated in many infectious and inflammatory diseases. In addition, there is no universally accepted cut-off value for its clinical use. [10]

Correa Fierro et al. (2025) used an untargeted LC-MS-based serum metabolomics approach to identify lipid biomarkers associated with dengue infection in children. Serum samples from RT-PCR-confirmed dengue patients and febrile controls were analyzed using UPLC-QTOF-MS, followed by multivariate analysis and lipid annotation. The findings should be interpreted with caution because metabolite levels are influenced by factors such as age, nutritional status, host metabolism, and other febrile illnesses. Most of the identified metabolites also require further experimental validation before they can be considered reliable biomarkers. [11]

Bernal et al. (2023) analyzed acute-phase serum samples from 145 confirmed dengue patients to evaluate serological markers, inflammatory mediators, and routine clinical parameters associated with severe dengue. The study identified age, comorbidities, anti-DENV IgG levels, thrombocytopenia, lymphopenia, and elevated chymase as significant biomarkers. The study is limited because several of these biomarkers can also be influenced by underlying health conditions and other infections, reducing their specificity for dengue. Larger studies are also needed to validate these findings. [12]

Suppiah et al. (2025) performed a pilot RNA-seq study on PBMC samples from dengue patients with warning signs and severe dengue in Malaysia to identify host transcriptomic biomarkers associated with infection and disease progression. Differential gene expression and pathway enrichment analyses identified several immune- and platelet-related genes, including IFI27, KCTD14, SDC1, CCL7, ACOD1, TNFSF15, and SERPINB2, as potential transcriptomic biomarkers. The study suggested that these genes could improve our understanding of the host response during dengue infection and serve as promising candidates for future biomarker validation.

Li-Min et al. (2021) used an integrated bioinformatics approach to analyze three GEO transcriptomic datasets (GSE28405, GSE38246, and GSE51808) to identify genes that were differentially expressed in dengue patients compared with healthy controls. The analysis identified 69 common differentially expressed genes, and functional enrichment showed that many of these genes were involved in type I interferon signaling, immune response, cell cycle regulation, DNA replication, and protein translation. Further protein interaction and miRNA

analyses highlighted IFI44L and IFI6 as the most promising transcriptomic biomarkers, suggesting that interferon-mediated immune responses play a key role in dengue pathogenesis.

2. Research Gap

Dengue is commonly diagnosed using clinical examination along with laboratory tests such as NS1 antigen detection, RT-PCR, and serological assays. Although these methods are widely used, each has its own limitations depending on the stage of infection. Researchers have also identified several promising biomarkers using proteomics, metabolomics, immunoassays, and flow cytometry. However, many of these biomarkers still require further validation and may be affected by other infections or biological factors. In addition, these approaches do not directly capture changes in host gene expression, which can provide valuable insights into the molecular response to dengue infection.

3. Research Motivation

When the body is infected with the dengue virus, the expression of many host genes changes as part of the immune response. Studying these changes can provide a better understanding of the molecular mechanisms involved in dengue infection and the host response to the virus. Identifying transcriptomic biomarkers may help reveal key genes and pathways associated with dengue, provide candidates for future validation studies, support the development of molecular diagnostic tools, and contribute to the identification of potential therapeutic targets.

4. Research Question

Which differentially expressed genes and enriched biological pathways are associated with dengue infection and may serve as potential transcriptomic biomarkers?

5. Objectives

- To identify differentially expressed genes (DEGs) associated with dengue infection.
- To identify key biological pathways involved in dengue infection.
- To identify candidate biomarker genes for future validation and development of targeted molecular diagnostic assays.

CHAPTER III

BIONFORMATICS DATABASES AND TOOLS USED

Phase 1: Data Collection and Quality Control

Databases used in this phase: NCBI and ENA

NCBI (National Center of Biotechnology Information) is a public database that stores a large collection of biological data, including DNA, RNA, protein sequences, and sequencing datasets. Researchers can search for datasets using keywords or accession numbers and download raw sequencing data from the Sequence Read Archive (SRA) for further analysis.

The European Nucleotide Archive (ENA) is a public database maintained by EMBL-EBI that stores nucleotide sequencing data and related information. It allows users to search for sequencing datasets using accession numbers, organism names, or study IDs and download the data in standard formats such as FASTQ.

Tools used in this phase: fastp and FastQC

FastQC is a quality control tool used to check the quality of raw sequencing data. It examines read quality, GC content, sequence length, duplication levels, adapter contamination, and other quality measures. It generates an HTML report that helps identify any problems in the sequencing data before analysis.

fastp is a preprocessing tool used to clean raw sequencing data. It removes adapter sequences, trims low-quality bases, filters out poor-quality reads, and corrects some sequencing errors. It also generates a quality report, ensuring that only high-quality reads are used for downstream analysis.

Phase 2: Alignment and Assembly of Sequencing Reads

Tool used in this phase: STAR

STAR (Spliced Transcripts Alignment to a Reference) is an RNA-seq aligner that efficiently handles spliced reads. It works in two main steps:

1. Seed searching:

STAR finds the longest exact matching parts of each read in the genome (called seeds or Maximal Mappable Prefixes (MMPs)). It keeps searching the remaining part of the read step by step until all parts are found. If matches are not perfect, it can extend them or remove bad-quality parts (soft clipping).

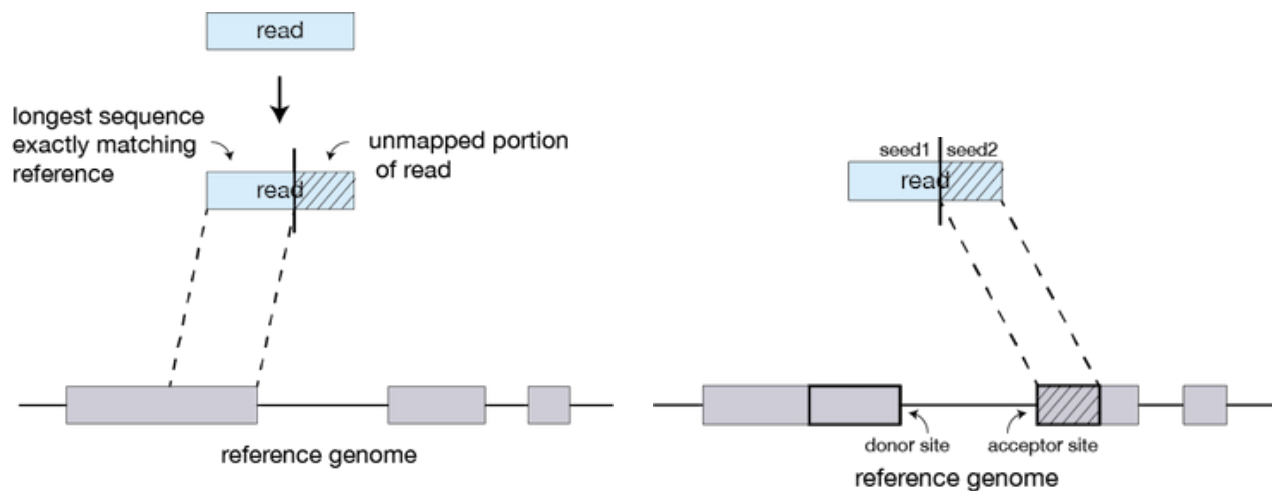


Figure 9: Sequential searching of only unmapped portions of reads [4]

2. Clustering, stitching, and scoring:

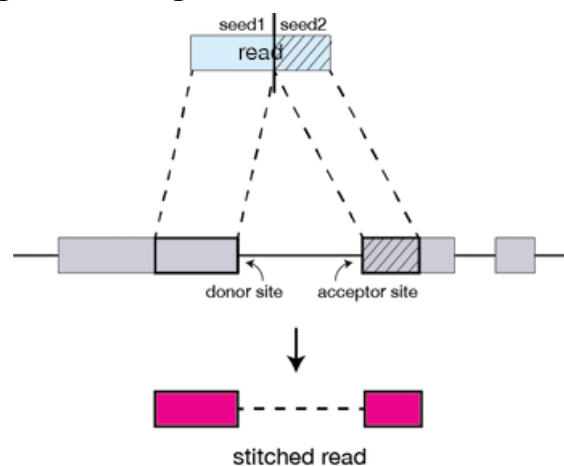


Figure 10: Clustering of seeds that are not multi-mapping [4]

STAR groups these matching parts based on their positions in the genome, then joins them to form a full alignment. It chooses the best match based on mismatches, gaps, and small insertions or deletions.

Phase 3: Quantification of Transcript Abundance

Tool used in this phase: featureCounts

featureCounts is a tool used in RNA-seq to count how many reads map to each gene or genomic feature. It checks where each read aligns on the genome and sees if it overlaps with a gene or feature by at least 1 base. If it overlaps, it is counted as a “hit.” Some reads may overlap more than one gene, and in RNA-seq these are often ignored because it is unclear which gene they came from.

To make the process fast, featureCounts first matches chromosome names using a hash table so reads and annotations can be quickly linked. The genome is then divided into large bins and smaller blocks inside them, so reads are first matched to bins, then blocks, and finally to genes. This step-by-step system makes the process efficient even for large genomes.

Phase 4: Normalization and Differential Expression modelling

Tool used in this phase: DESeq2

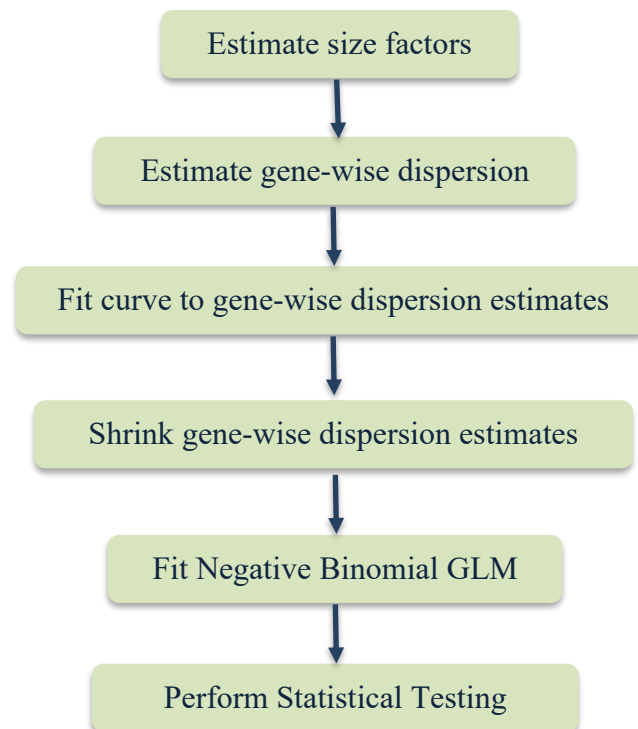


Figure 11: DESeq2 Steps

In DESeq2, the first step is estimating size factors, which normalizes count data by accounting for differences in sequencing depth and library size across samples using the median-of-ratios method. Next, dispersion estimation measures the biological variability in gene expression beyond random sampling noise. DESeq2 then calculates gene-wise dispersion estimates for each gene individually, followed by fitting a mean–dispersion relationship to model how dispersion changes with average expression levels.

Using this trend, final dispersion estimates are obtained through shrinkage, making them more stable and reliable. Finally, DESeq2 fits a negative binomial model and performs statistical testing to identify genes that are significantly differentially expressed between experimental conditions.

Phase 5: GO and Pathway Analysis

Tool used in this phase: clusterProfiler package in R

In this phase, the clusterProfiler package was used to understand what the differentially expressed genes actually do in the body and which biological systems they may affect.

GO (Gene Ontology) analysis is based on the Gene Ontology database, which is a structured database that describes gene functions in a consistent way across three areas: Biological Process (what the gene is involved in, like immune response or cell division), Molecular Function (what the gene does, such as binding or enzyme activity), and Cellular Component (where it works in the cell, like the nucleus or membrane). It checks whether any of these GO terms appear more often in the gene list than expected by chance, which suggests they may be important in the condition being studied.

KEGG pathway analysis uses the KEGG (Kyoto Encyclopedia of Genes and Genomes) database, which contains well-curated biological pathways. These pathways show how genes work together in networks involved in processes like immune signaling, metabolism, and disease mechanisms. By mapping genes to these pathways, it becomes easier to see which biological pathways may be affected in the condition.

More generally, pathway analysis methods can be grouped into three main types. Over-Representation Analysis (ORA) takes a list of significant genes and checks whether any known pathway contains more of these genes than expected by chance. Gene Set Enrichment Analysis (GSEA) uses all genes ranked by expression change and looks for pathways where genes tend to cluster at the top or bottom of the list, helping detect small but coordinated changes. Topology-based analysis goes a step further by also considering how genes interact within a pathway, including network structure and relationships, to give a more detailed view of how the pathway is affected.

CHAPTER IV

METHODOLOGY

1. Dataset collection

The RNA-seq dataset PRJNA1279769 was downloaded from NCBI. It contains whole transcriptome sequencing data from 60 NS1 antigen-positive dengue patients and 20 NS1 antigen-negative febrile illness controls admitted to Max Super Specialty Hospital, Delhi, India. The samples represent a range of disease severity from non-severe to severe dengue infection. RNA was extracted from whole blood samples, and after rRNA depletion, libraries were prepared using the TruSeq Stranded Total RNA Library Prep Kit.

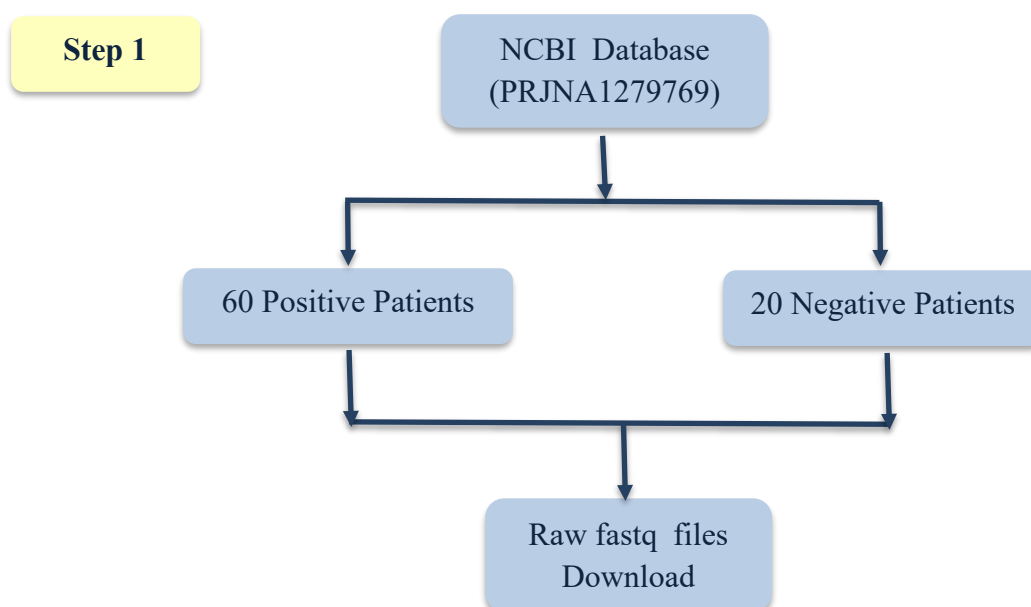


Figure 12: Dataset Collection steps

2. Quality Control

The FASTQ paired-end files were downloaded from ENA using terminal curl commands. After downloading, the raw reads were preprocessed using the tool fastp which performs adapter removal, quality trimming, and read filtering in a single step. It automatically detects adapters, so manual specification is not required.

The process was run using 2 threads on a 32-thread workstation. The output consisted of cleaned FASTQ files along with reports on read quality and sequencing depth. After this, the cleaned reads were further evaluated using FastQC to confirm that adapters were removed and that the reads passed quality checks. These cleaned FASTQ files were then used for downstream analysis.

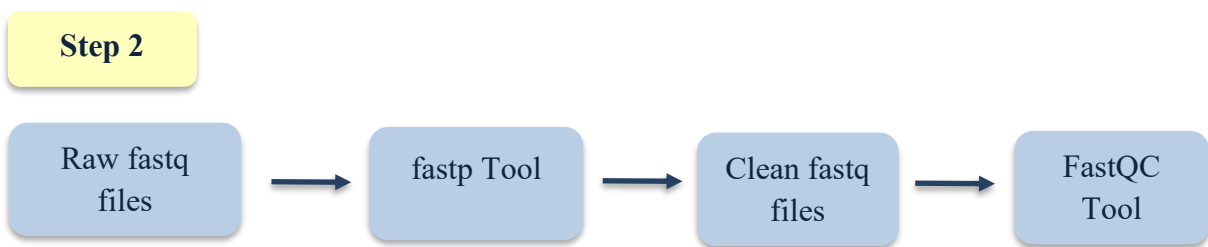


Figure 13: Quality Control steps

3. Reference Genome Indexing

The human reference genome was downloaded from NCBI as GCF_000001405.40_GRCh38.p14_genomic.fna, along with the annotation file GCF_000001405.40_GRCh38.p14_genomic.gtf. The .fna file is a FASTA nucleotide sequence file that contains the reference genome, and the .gtf file (Gene Transfer Format) contains the gene annotation information such as gene, exon, and transcript locations. Genome indexing was performed using STAR with 28 threads (out of 32) using the following parameters:

- --runThreadN → number of threads used (28 out of 32 used)
- --runMode genomeGenerate → builds the genome index
- --genomeDir → output directory for storing index files
- --genomeFastaFiles → reference genome FASTA (.fna) file
- --sjdbGTFfile → gene annotation file (.gtf)
- --sjdbOverhang → read length minus 1 (improves splice junction detection) (default value of 100 was used)

Output files generated by STAR indexing:

- Genome → main binary genome index used for alignment
- SA / SAindex → suffix array files for fast searching
- genomeParameters.txt → parameters used for genome generation
- Log.out → log file showing indexing process details
- chrName.txt / chrLength.txt / chrStart.txt → chromosome information files
- exonInfo.tab / exonGeTrInfo.tab → exon-level annotation information
- geneInfo.tab → gene-level annotation summary
- transcriptInfo.tab → transcript-level annotation information
- sjdbInfo.txt → splice junction database information
- sjdbListFromGTF.out.tab → splice junctions extracted from GTF
- sjdbList.out.tab → final curated splice junction list used for alignment

These index files allow STAR to quickly and accurately map RNA-seq reads to the reference genome, especially across splice junctions.

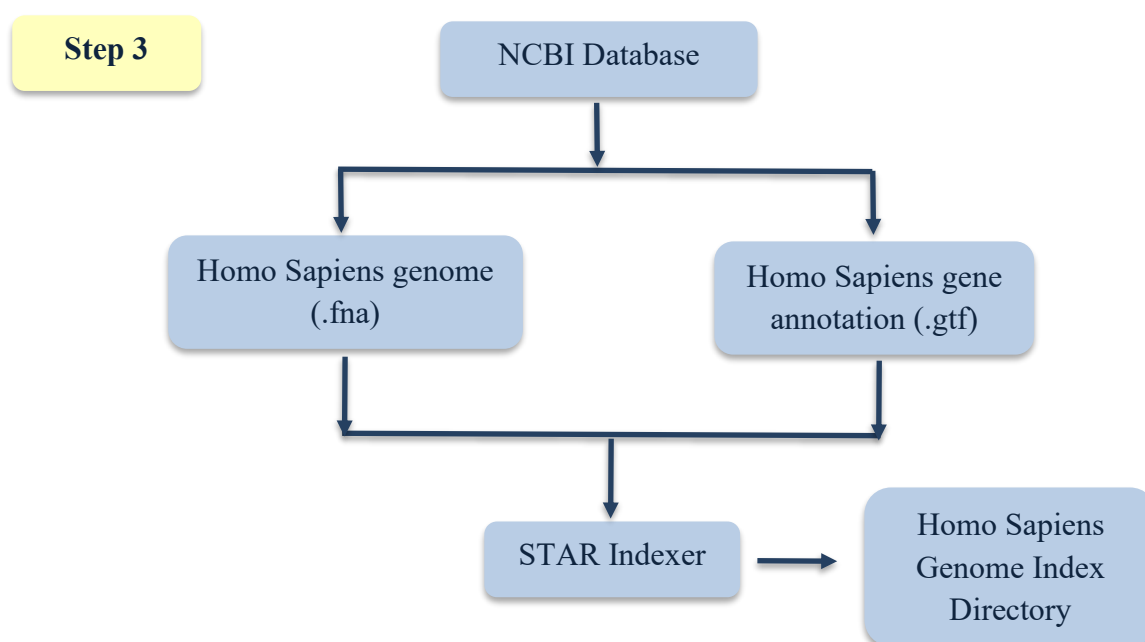


Figure 14: Genome Indexing steps

4. Aligning to the Indexed Reference Genome

Each of the clean FASTQ files were aligned to the indexed human genome using **STAR**, which produces a BAM file for each sample. Aligning was performed using STAR with 28 threads (out of 32) using the following parameters:

- `--runThreadN` → number of CPU threads used (28)
- `--readFilesIn` → path to input FASTQ files
- `--genomeDir` → directory containing STAR genome index files
- `--outFileNamePrefix` → prefix for all output files
- `--outSAMtype BAM SortedByCoordinate` → output aligned reads in sorted BAM format

After alignment, the results are stored in BAM format, which records how each read maps to the reference genome. A SAM file is a text-based format that contains alignment details for each read, including its name, reference chromosome, mapping position, mapping quality, CIGAR (Compact Idiosyncratic Gapped Alignment Report) string, nucleotide sequence, and quality scores, along with optional information such as pairing and alignment flags. A BAM file contains the same information but in a compressed binary format, making it more storage-efficient and faster for downstream processing, though not directly human-readable.

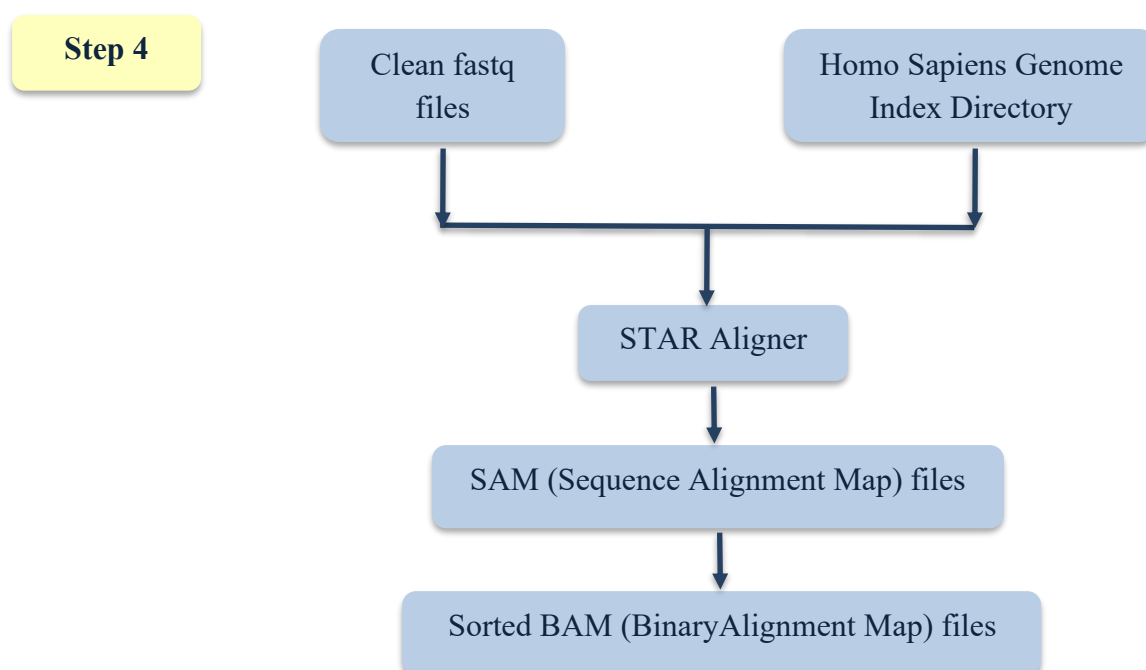


Figure 15: Alignment steps

5. Quantification of Reads

The aligned BAM files were used for gene quantification using featureCounts, along with the human GTF annotation file. featureCounts scans each BAM file and identifies overlaps between aligned reads and annotated gene regions. Reads that overlap a gene are counted to generate gene-level expression values. The output for each sample is saved as a separate file, and all sample-level outputs were later combined to form a final gene count matrix. In this matrix, rows represent genes and columns represent samples, and is used for downstream analysis.

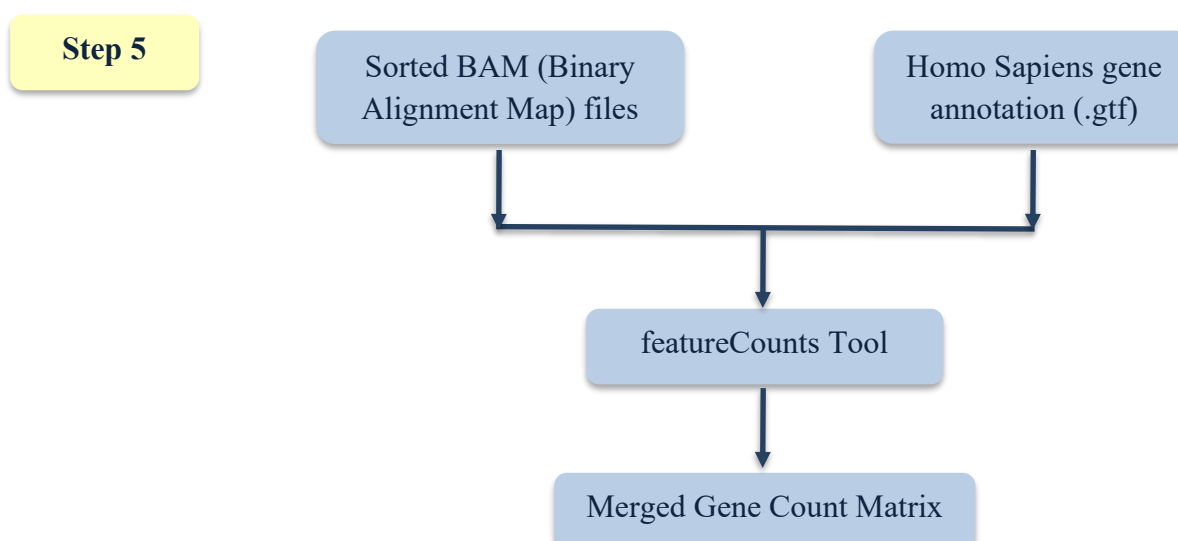


Figure 16: Quantification steps

6. Differential Expression Analysis

The gene count matrix from featureCounts was imported into R for further analysis. A metadata file with sample information (Severe, Non-Severe, and Negative groups) was also loaded to define the study design. Differential expression analysis was done using the DESeq2 package. A DESeq2 dataset was created using the count data and metadata, and the group information was used for comparison. Genes with very low counts were removed to improve accuracy.

The data was then processed using DESeq2, which performs normalization, dispersion estimation and statistical testing to find differentially expressed genes. Variance stabilizing transformation (VST) was used to convert the data into a normalized form for analysis. For each gene, values like base mean expression, log2 fold change, p-value, and adjusted p-value were calculated. Different plots were also created to visualize the results.

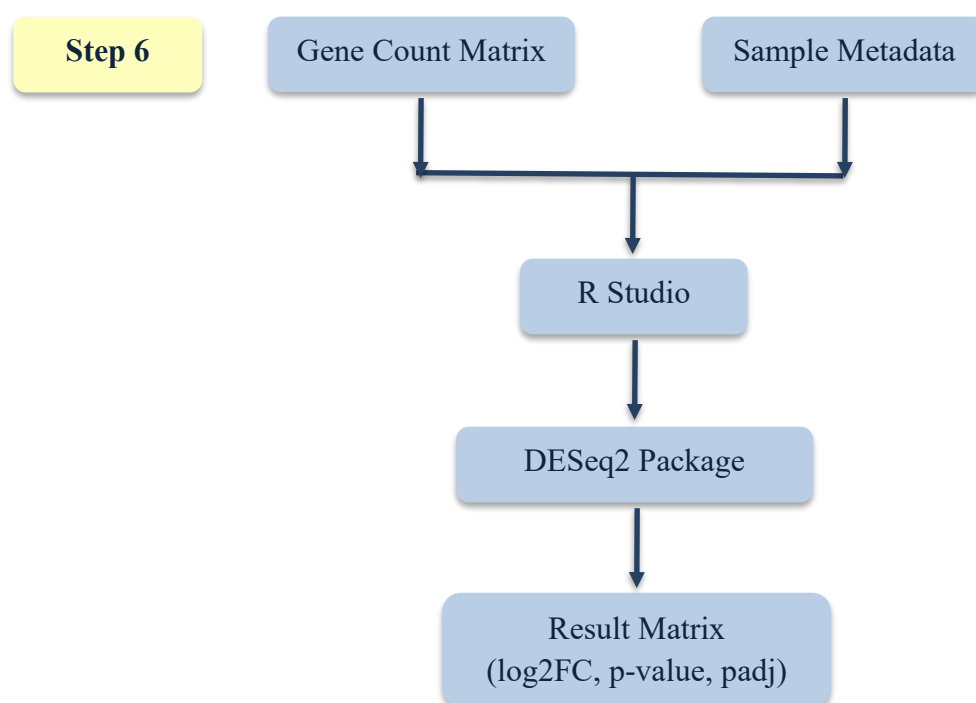


Figure 17: Differential Expression Analysis steps

7. Gene Ontology (GO) Enrichment Analysis

The significantly differentially expressed genes identified from DESeq2 were used for Gene Ontology (GO) enrichment analysis. This was performed using the enrichGO function from the clusterProfiler package. The analysis was carried out with adjusted p-value cutoff of 0.05, a minimum gene set size (minGSSize) of 5 and a maximum gene set size (maxGSSize) of 500 to ensure reliable and biologically meaningful terms. After enrichment, the GO terms were further filtered to retain only those related to immune response and dengue infection.

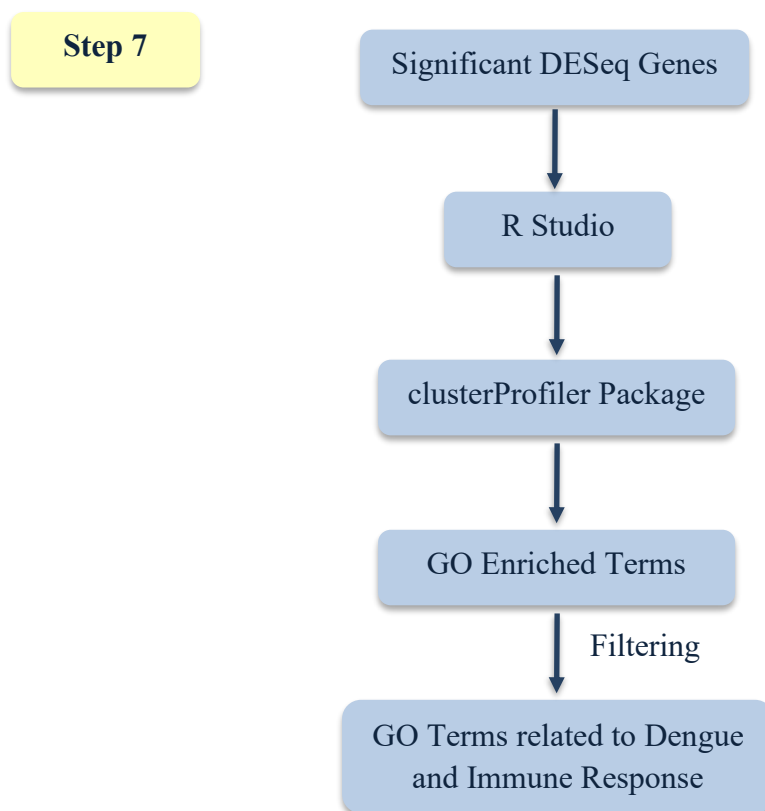


Figure 18: Gene Ontology (GO) Enrichment steps

8. KEGG Pathway Analysis

The significantly differentially expressed genes identified from DESeq2 were used for KEGG pathway enrichment analysis. This was performed using the `enrichKEGG` function from the `clusterProfiler` package with *Homo sapiens* as the reference organism. The analysis was carried out using an adjusted p-value cutoff of 0.05, a minimum gene set size (`minGSSize`) of 5, and a maximum gene set size (`maxGSSize`) of 500 to identify significantly enriched pathways. After enrichment, pathways related to immune response and dengue pathogenesis were selected for further interpretation.

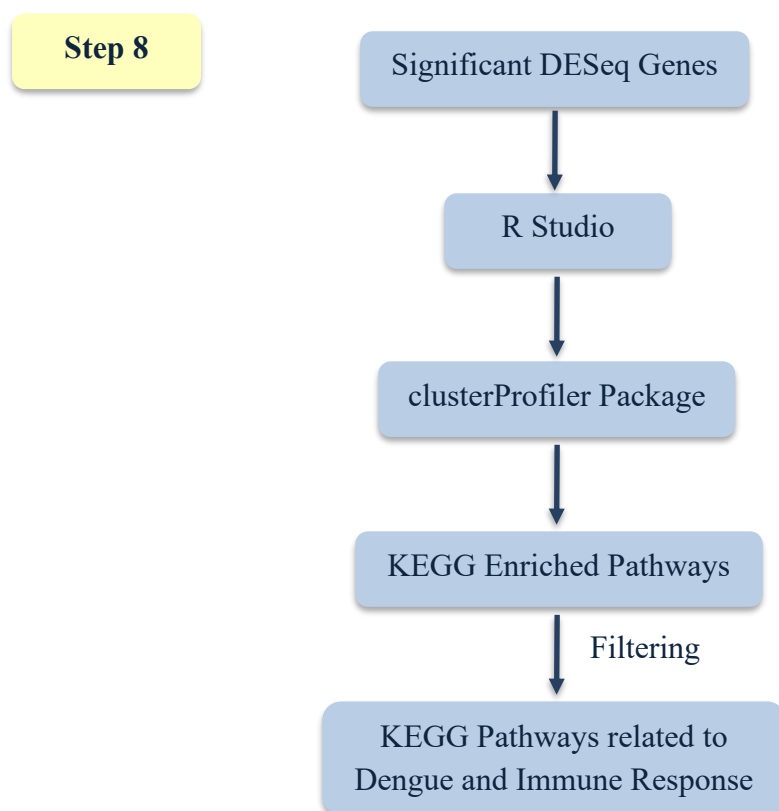


Figure 19: KEGG Pathway Analysis steps

9. Visualization of Results and Application Development

Different plots were created to visualize and understand the results of the analysis. To make the results easier to explore, a web application was developed using Shiny R. All important results, including gene expression patterns, selected genes, and model outputs, were included in the app. The application can be accessed at: <https://varshas.shinyapps.io/Denguscope/>.

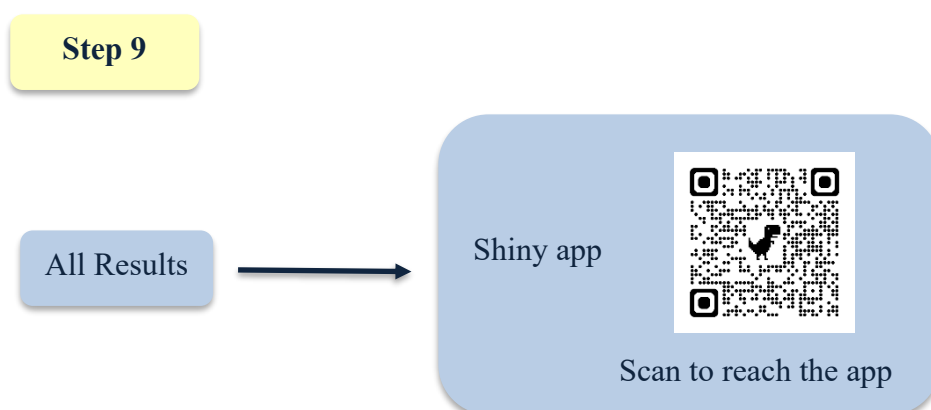


Figure 20: R Shiny App Development

CHAPTER V

RESULTS

1. Sample Selection

Out of the 80 samples in the dataset, only 73 samples were used for further analysis. The remaining 7 samples were excluded as their FASTQ files were not of sufficient quality for processing.

2. Data Distribution

To get an overview of the sample distribution, pie charts were created to show the distribution of samples based on gender, age, platelet count ($10^3/\mu\text{L}$), and sample type (Negative, Non-Severe, and Severe).

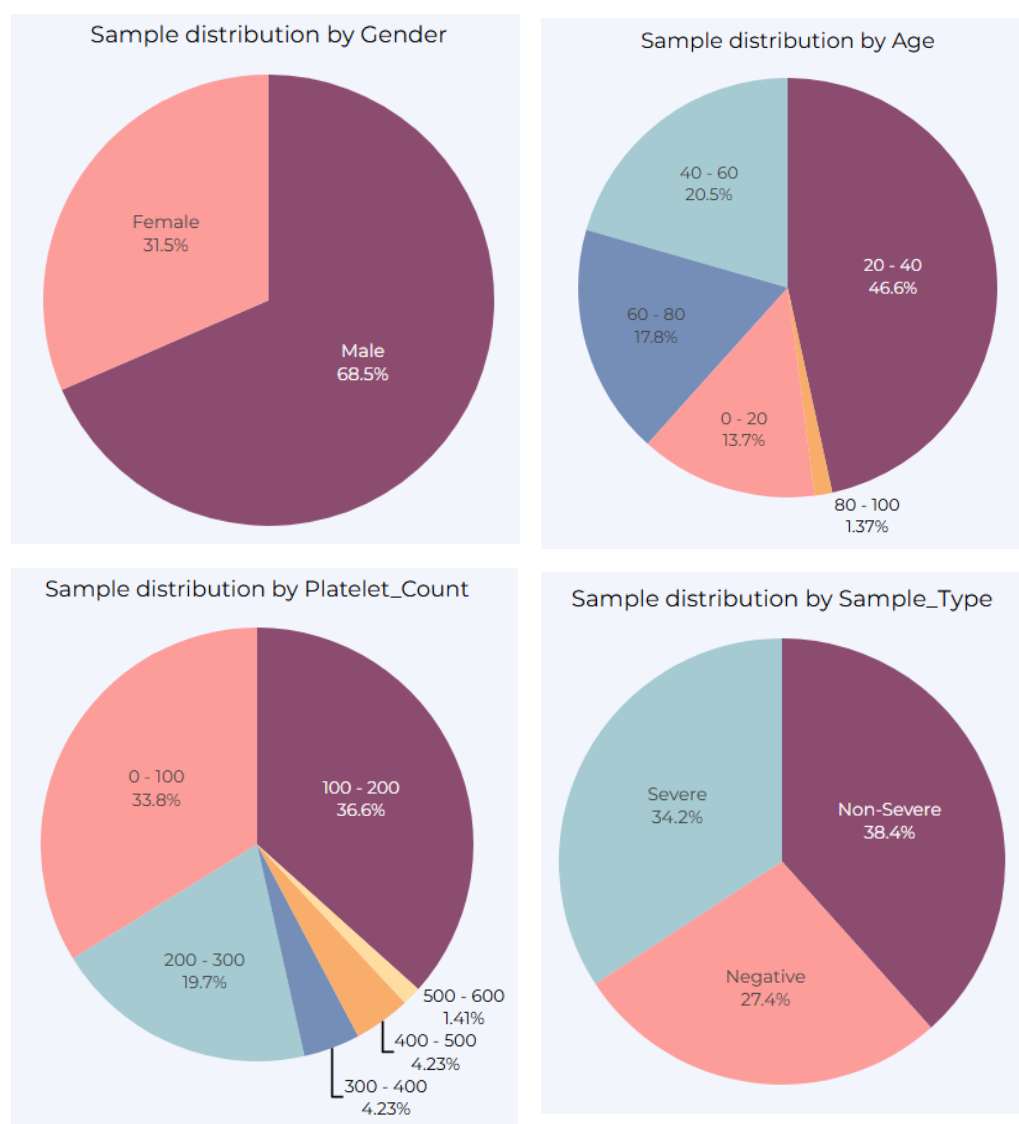


Figure 21: Sample Distribution based on gender, age, platelet count ($10^3/\mu\text{L}$), and sample type

1. Principal Component Analysis (PCA)

A Principal Component Analysis (PCA) plot was generated to examine the overall variation in gene expression across the three sample groups: Negative, Non-Severe, and Severe. PCA reduces the complexity of the data, making it easier to identify patterns and similarities among samples. The plot showed a clear separation between the negative samples and the positive samples. Most Negative samples clustered on the positive side of the first principal component (PC1), while the Non-Severe and Severe samples were mainly distributed on the opposite side. Although there was some overlap between the Non-Severe and Severe groups, both groups were clearly separated from the Negative samples.

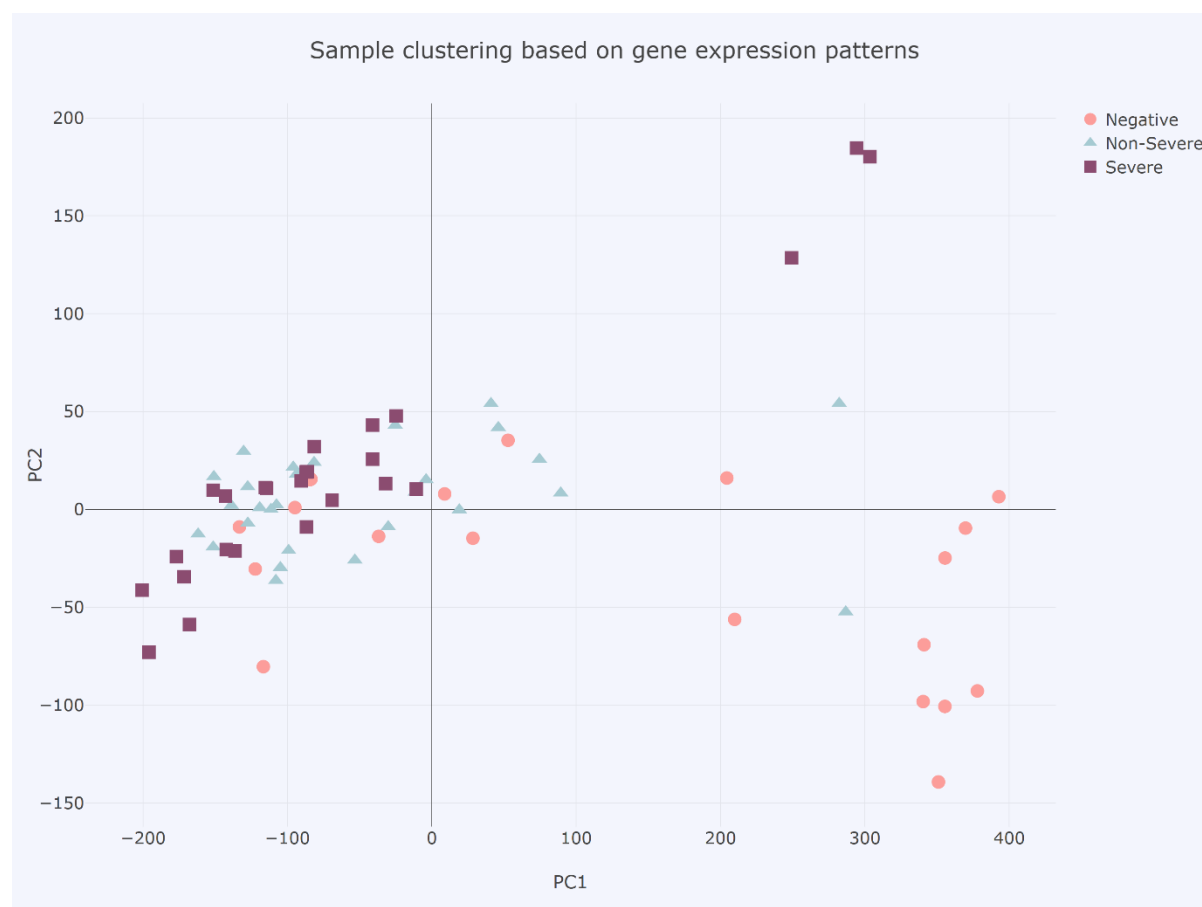


Figure 22: Principal Component Analysis (PCA) plot

2. MA Plot Analysis

An MA (Mean - Average) plot was generated to visualize the relationship between the average expression level of genes and their $\log_2$ fold change between the two groups. Most genes were clustered around a $\log_2$ fold change of zero, indicating similar expression levels across the groups. A smaller number of genes showed significant upregulation, while a larger number exhibited significant downregulation, suggesting that the disease condition was associated with more widespread decreases than increases in gene expression.

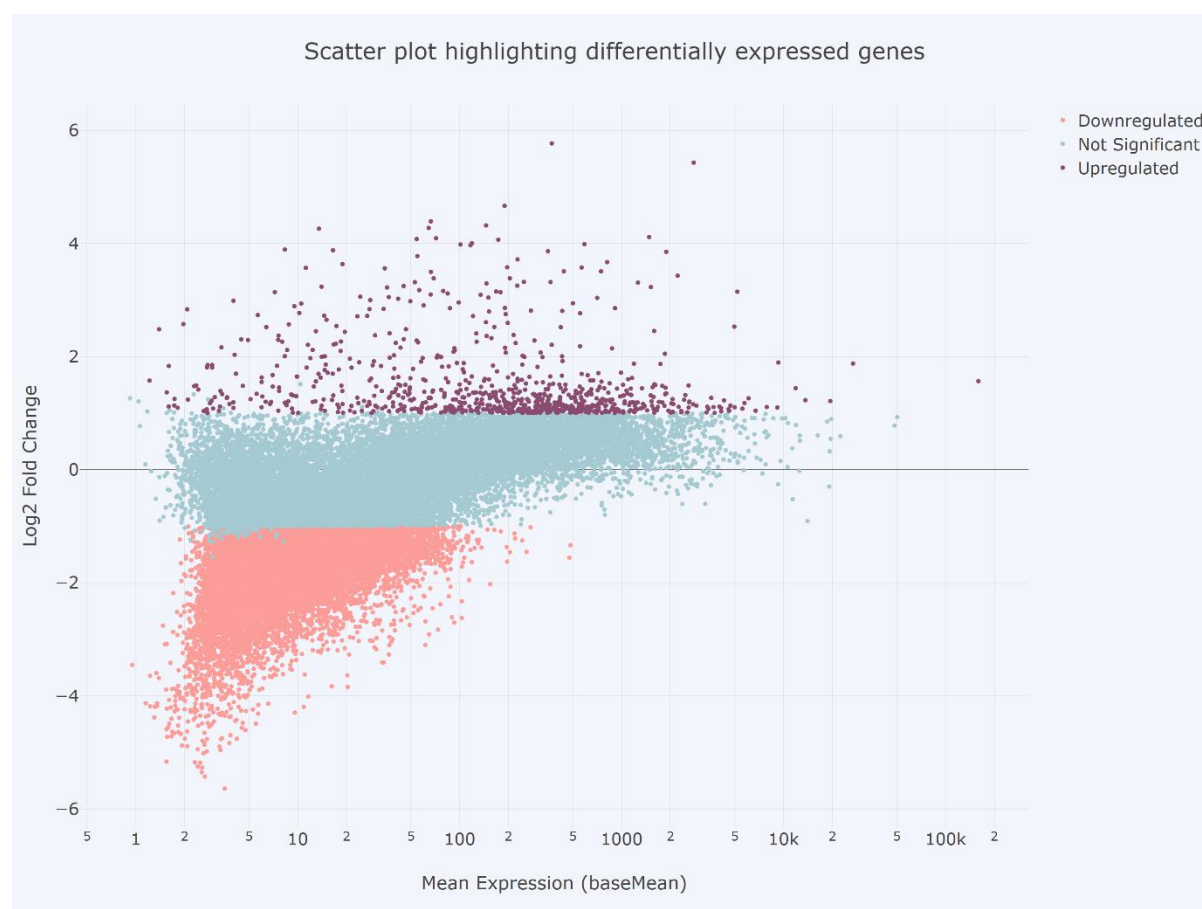


Figure 23: MA Plot of Differential Gene Expression

3. Gene Expression Analysis

Boxplots were generated to visualize the expression levels of different genes across the three groups: Negative, Non-Severe, and Severe. These plots enabled comparison of gene expression between the groups and helped identify genes with distinct expression patterns. Gene expression was also visualized across all individual samples to assess sample-to-sample variation, providing an overview of the distribution and consistency of expression levels throughout the dataset.

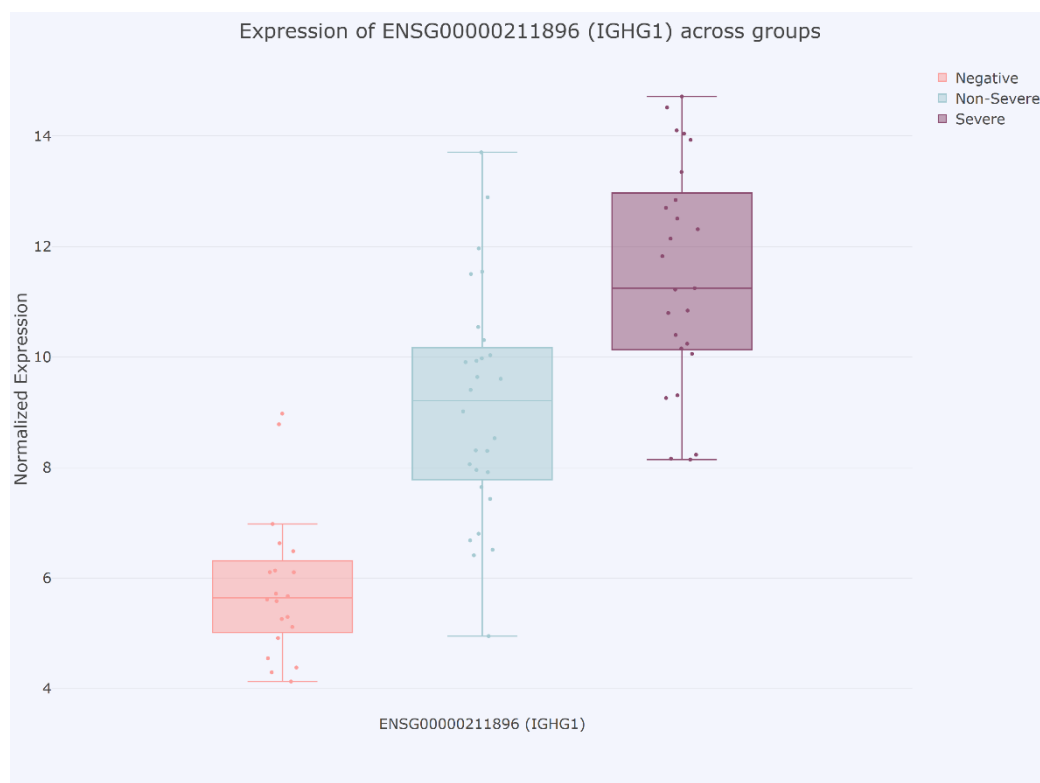


Figure 24: Gene Expression of the Top Upregulated Gene IGHG1 across groups

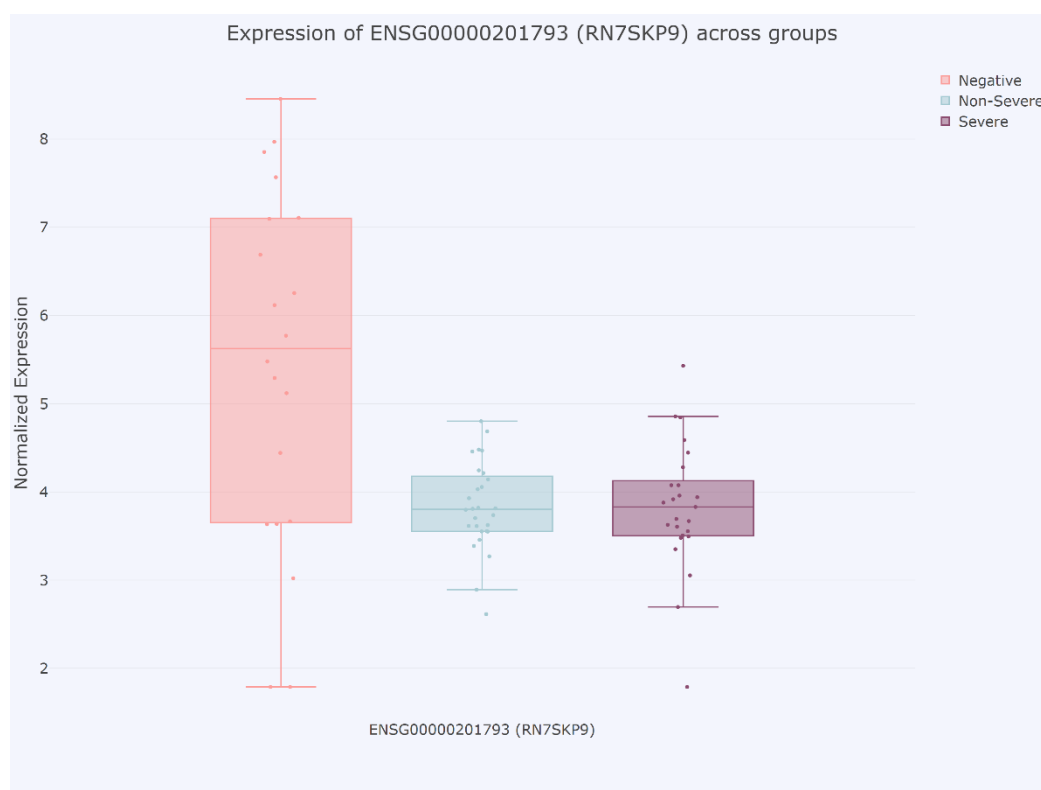


Figure 25: Gene Expression of the Top Downregulated Gene RN7SKP9 across groups

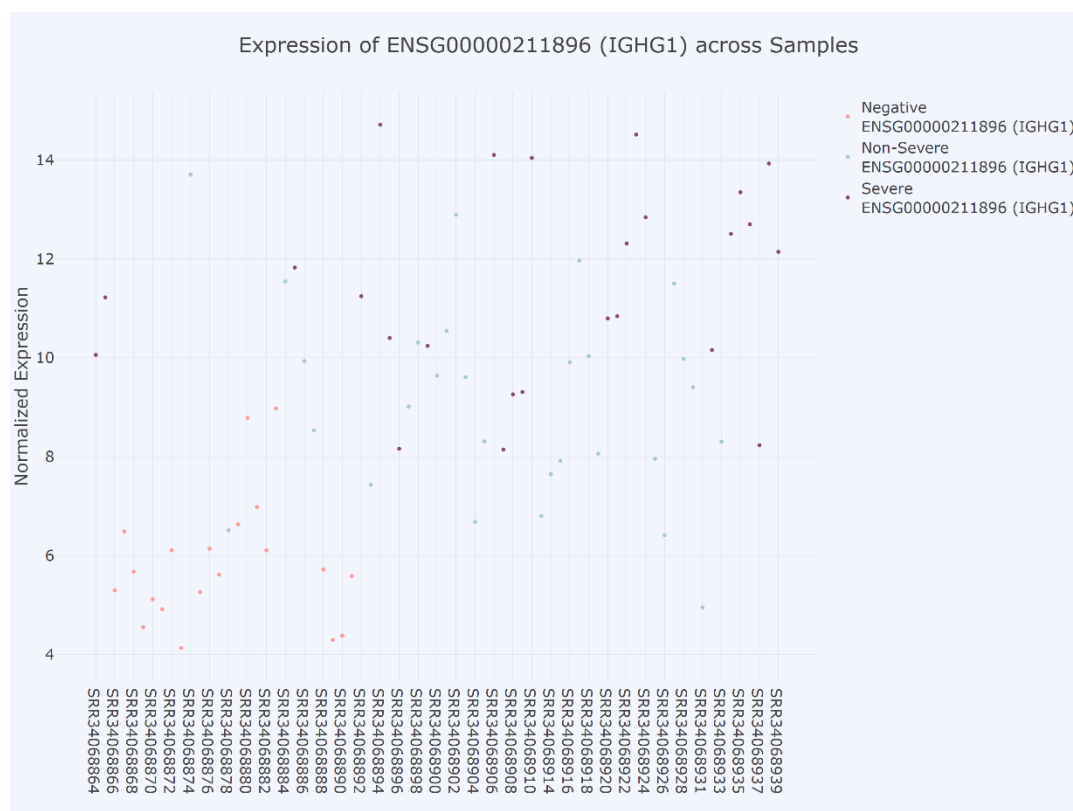


Figure 26: Gene Expression of the Top Upregulated Gene IGHG1 across samples

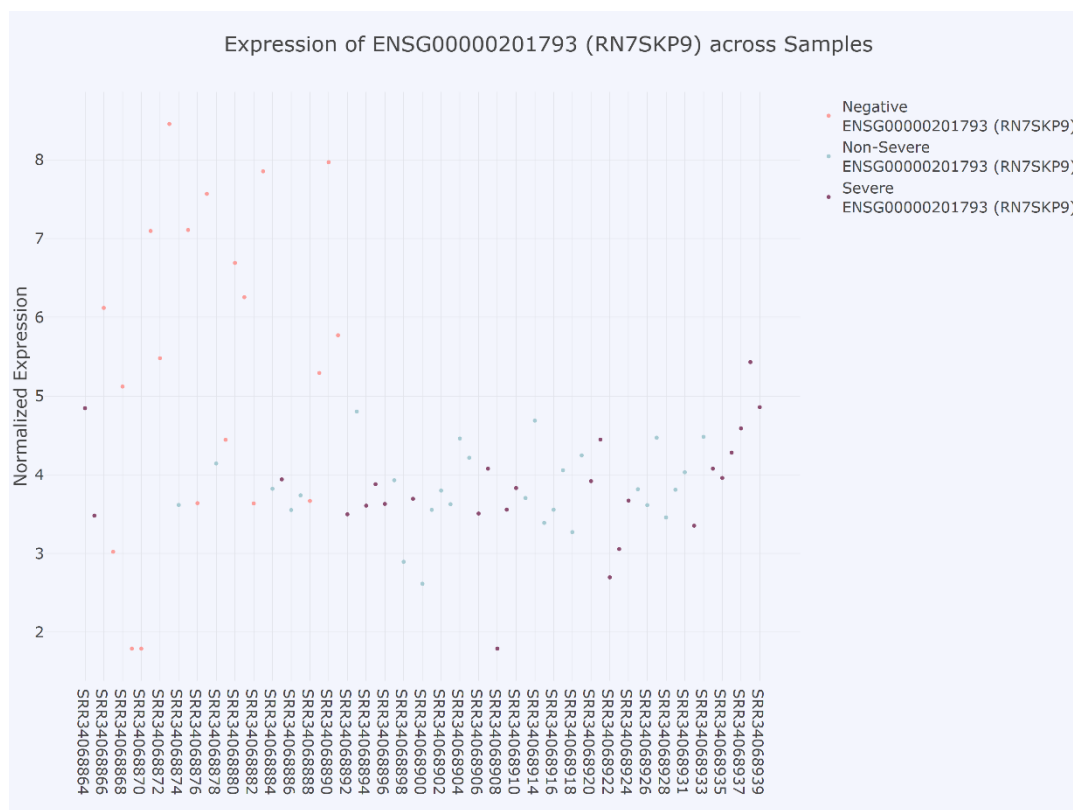


Figure 27: Gene Expression of the Top Downregulated Gene RN7SKP9 across samples

7. Differential Gene Expression Analysis

Genes were considered differentially expressed if they had an adjusted p-value (FDR) < 0.05 and an absolute $\log_2$ fold change > 1. Using these criteria, significantly upregulated and downregulated genes were identified, as summarized below.

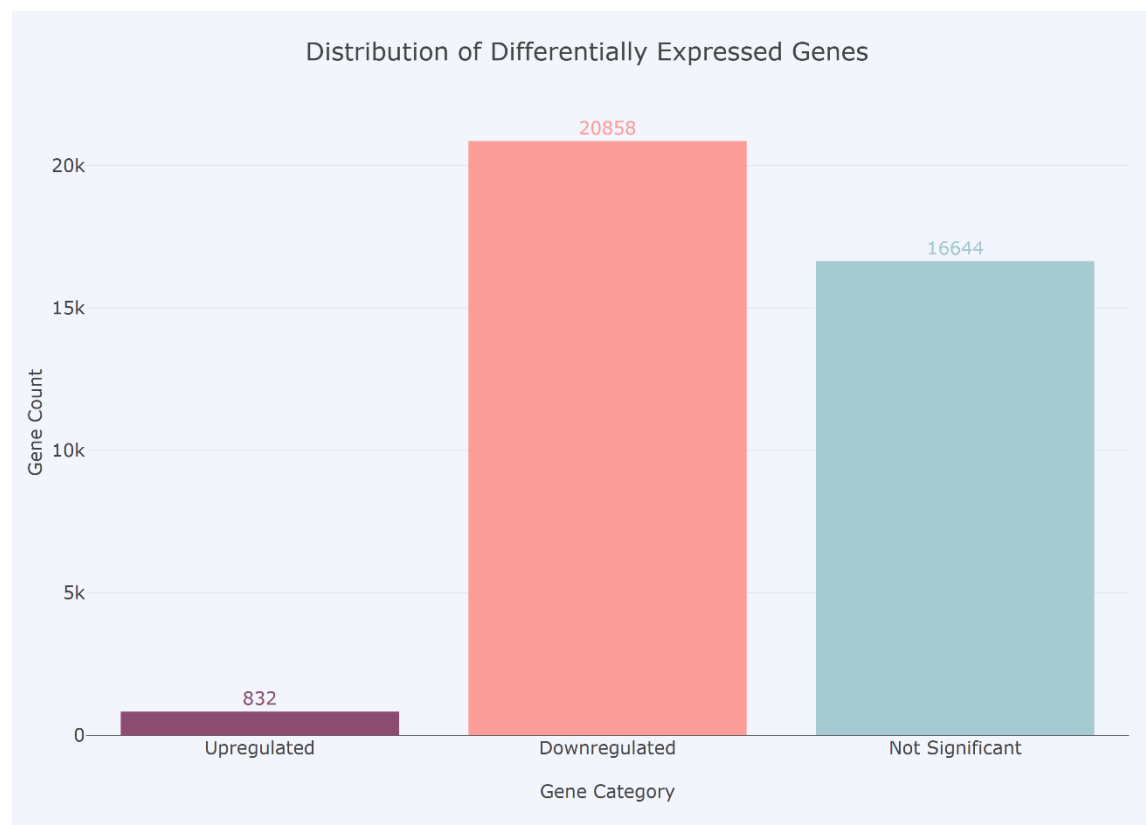


Figure 28: Distribution of Differentially Expressed Genes

Ensembl ID	Gene Symbol	log2FoldChange	padj	baseMean
ENSG00000211896	IGHG1	5.429808607	1.09E-23	2785.99
ENSG00000211662	IGLV3-21	5.769842411	9.93E-23	370.4928
ENSG00000165949	IFI27	3.852420964	3.65E-18	1885.493
ENSG00000273983	H3C8	3.981379645	4.72E-16	101.1707
ENSG00000004799	PDK4	2.362796867	5.05E-16	146.8294
ENSG00000148773	MKI67	2.943330147	1.15E-14	500.148
ENSG00000211648	IGLV1-47	4.390443999	3.09E-14	66.43761
ENSG00000277775	H3C7	3.383024321	5.24E-14	204.0753

ENSG00000211677	IGLC2	4.114802417	1.30E-13	1478.914
ENSG00000132465	JCHAIN	3.575434071	3.70E-13	566.0983
ENSG00000211962	IGHV1-46	4.274206057	9.57E-13	64.38995
ENSG00000211660	IGLV2-23	3.970126566	1.68E-12	116.5307
ENSG00000139514	SLC7A1	1.372954439	2.19E-12	167.4014
ENSG00000184357	H1-5	3.508739949	2.53E-12	439.7301
ENSG00000211653	IGLV1-40	4.319188939	3.62E-12	145.5743
ENSG00000286522	H3C2	3.321214132	4.16E-12	249.0402
ENSG00000236552	RPL13AP5	2.54075505	4.16E-12	17.33532
ENSG00000211659	IGLV3-25	3.777080787	5.46E-12	54.80336
ENSG00000101057	MYBL2	2.409284745	5.55E-12	126.6821
ENSG00000136938	ANP32B	1.523726521	5.80E-12	256.5365

Table 1: Top 20 Significant Upregulated Genes

Ensembl ID	Gene Symbol	log2FoldChange	padj	baseMean
ENSG00000201793	RN7SKP9	-3.13937	5.49E-18	29.77148
ENSG00000070985	TRPM5	-3.4084	8.66E-16	33.91835
ENSG00000284395	PERCC1	-3.86945	1.32E-14	6.049114
ENSG00000125508	SRMS	-3.0389	8.01E-13	16.65583
ENSG00000177106	EPS8L2	-2.22074	9.57E-13	24.26944
ENSG00000167549	CORO6	-2.34649	1.63E-12	21.07873
ENSG00000088256	GNA11	-2.29371	3.62E-12	26.40282
ENSG00000184916	JAG2	-3.021	3.96E-12	27.99487
ENSG00000287431	RENO1	-3.04193	4.18E-12	9.346243
ENSG00000161914	ZNF653	-1.91314	5.65E-12	16.61756
ENSG00000197599	CCDC154	-2.39249	5.87E-12	18.08841
ENSG00000187609	EXD3	-1.99291	9.80E-12	54.48831
ENSG00000130711	PRDM12	-3.74986	1.51E-11	5.167148
ENSG00000187583	PLEKHN1	-2.86615	1.51E-11	12.55039
ENSG00000183479	TREX2	-3.16961	1.72E-11	4.221063
ENSG00000158246	TENT5B	-2.96208	1.76E-11	7.463179
ENSG00000076826	CAMSAP3	-2.77823	2.08E-11	22.81149

ENSG00000130702	LAMA5	-2.70416	2.08E-11	92.52321
ENSG00000179943	FIZ1	-2.00486	2.17E-11	47.475
ENSG00000162006	MSLNL	-4.0128	2.19E-11	11.62545

Table 2: Top 20 Significant Downregulated Genes

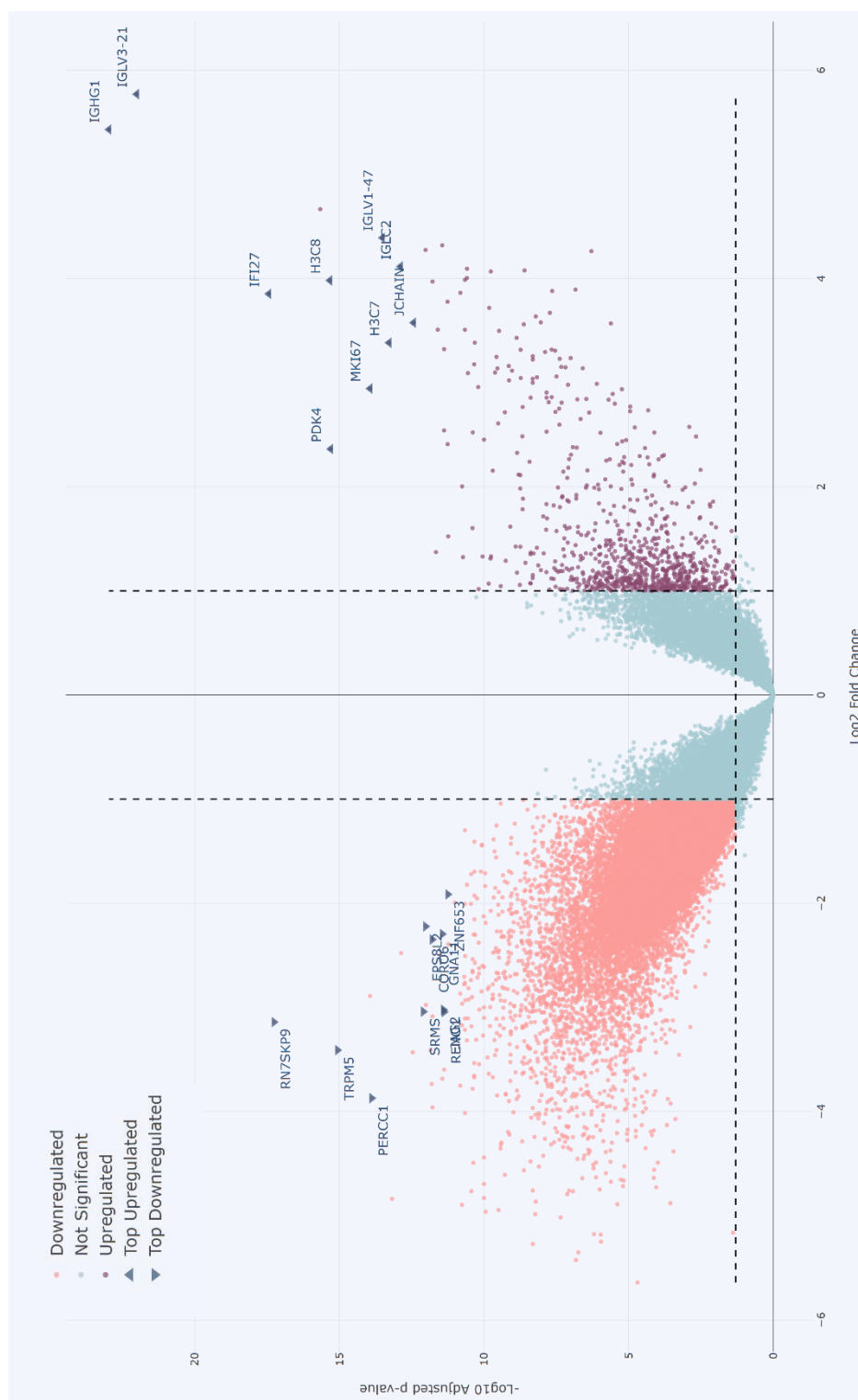


Figure 29: Volcano Plot of Differentially Expressed Genes

A volcano plot was generated using an adjusted p-value cutoff of 0.05 and a $\log_2$ fold change cutoff of 1 to visualize differential gene expression. The plot clearly distinguished significantly upregulated and downregulated genes from non-significant genes. The top 10 upregulated and top 10 downregulated genes identified from the analysis are presented above.

8. Heatmap of Top Differentially Expressed Genes

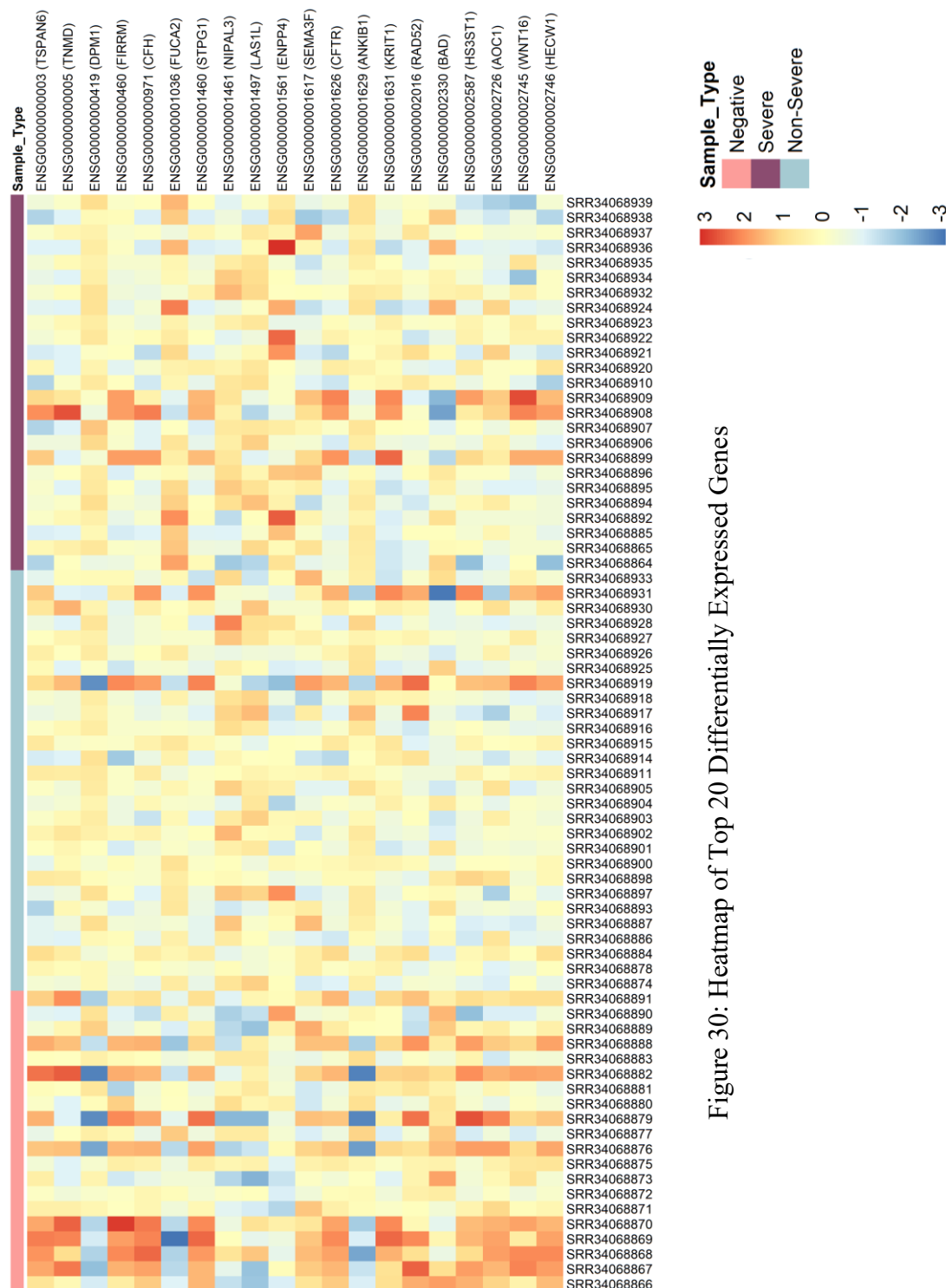


Figure 30: Heatmap of Top 20 Differentially Expressed Genes

A heatmap was generated using the top 20 differentially expressed genes selected based on an adjusted p-value (FDR) cutoff of 0.05. The heatmap visualizes the expression patterns of these genes across all samples, highlighting distinct differences between the Negative, Non-Severe, and Severe groups.

9. GO and KEGG Enrichment Analysis

The top enriched Gene Ontology (GO) terms from the Biological Process (BP), Cellular Component (CC), and Molecular Function (MF) categories, along with the top enriched KEGG pathways, were visualized using dot plots. In these plots, the size of each dot represents the number of genes associated with the term or pathway, while the color indicates its statistical significance. Together, these analyses provide an overview of the key biological functions, immune-related processes, and molecular pathways enriched in the dataset.

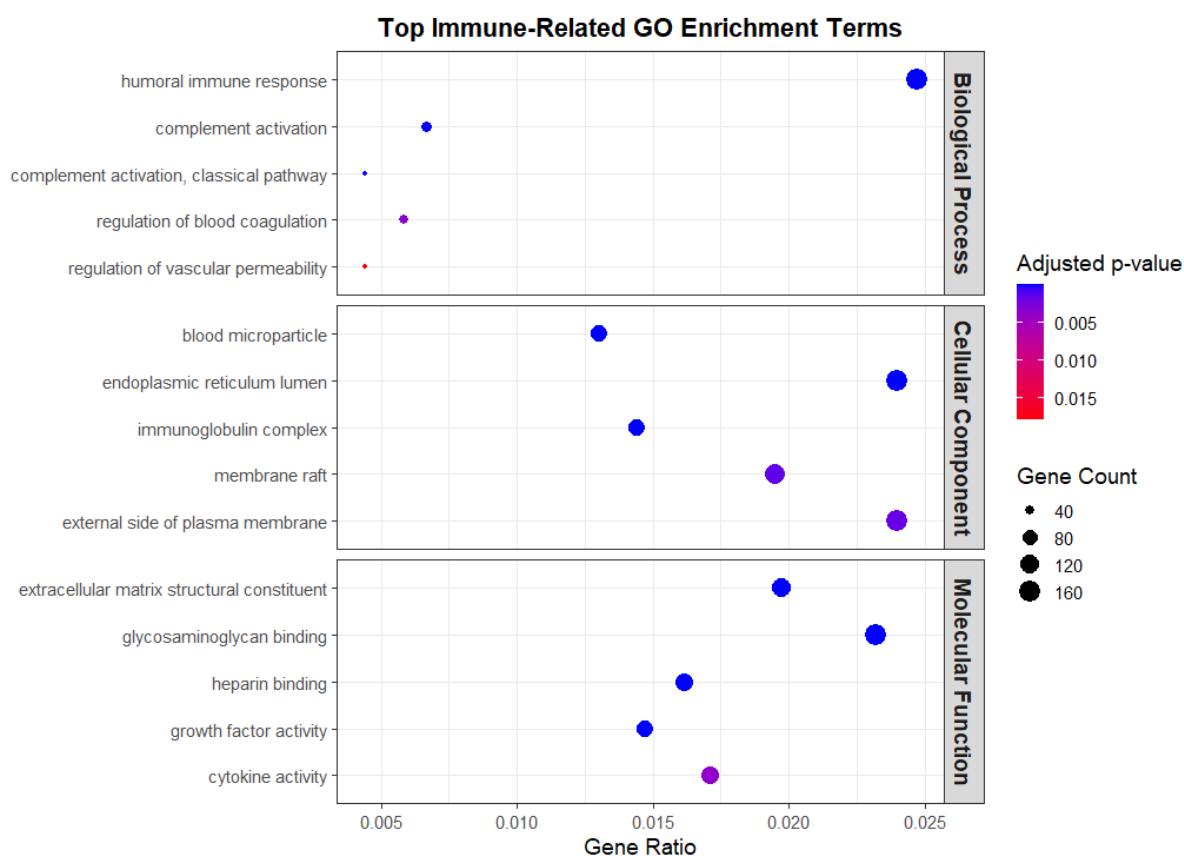


Figure 31: Dot Plot of Immune related Gene Ontology Terms

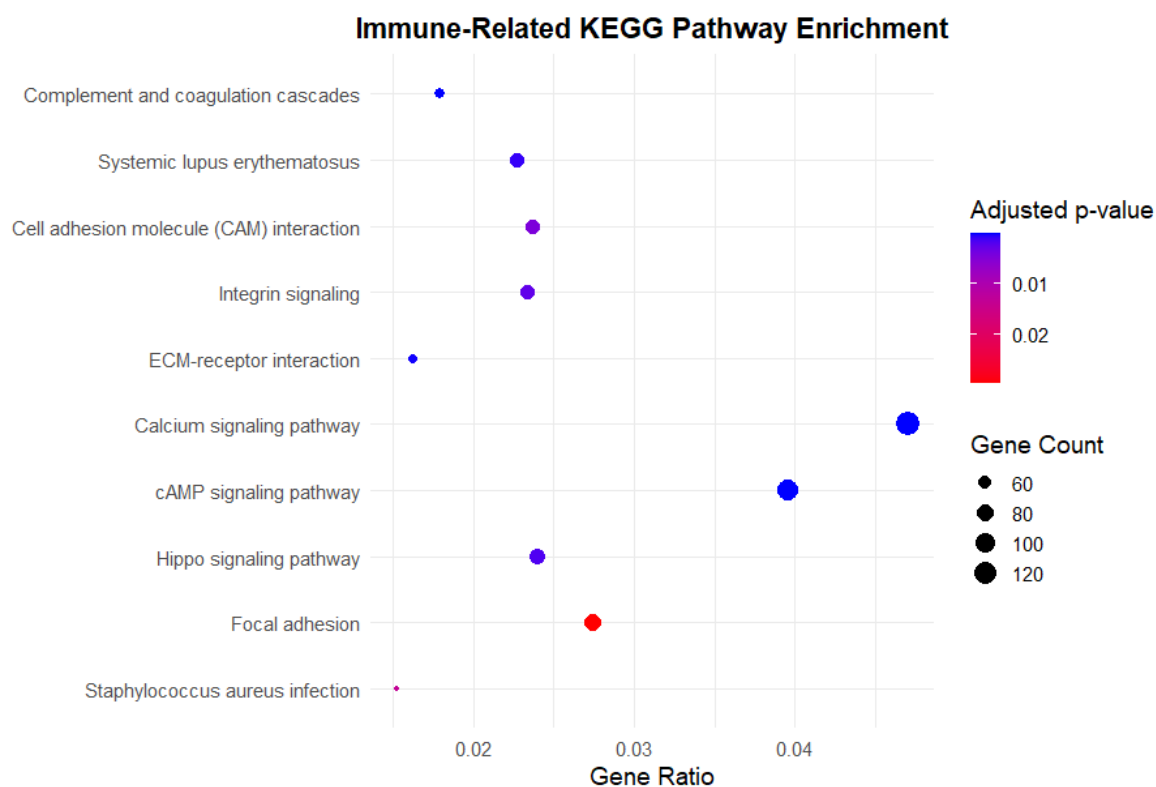


Figure 32: Dot Plot of Immune related KEGG Pathways

10. App Analysis

All of these results can also be explored interactively through the Shiny app. Users can modify the adjusted p-value (FDR) cutoff, $\log_2$ fold change cutoff, and the number of genes displayed, allowing them to explore differential expression and enrichment results under different filtering criteria.



Figure 33: Shiny App Homepage

DISCUSSION

This study identified significant changes in gene expression associated with dengue infection, demonstrating extensive alterations in the host response during disease. While several genes were significantly upregulated, a larger number were downregulated, indicating that dengue infection not only activates immune responses but also affects normal cellular functions.

Among the most highly upregulated genes, **IGHG1**, **IGLV3-21**, **IGLC2**, and **JCHAIN** are involved in antibody production and immunoglobulin function, suggesting strong activation of the humoral immune response. The increased expression of **IFI27**, an interferon-stimulated gene, indicates activation of antiviral defense mechanisms, while the upregulation of **MKI67** and **MYBL2** suggests enhanced immune cell proliferation and activity during infection.

Several downregulated genes, including **LAMA5**, **JAG2**, **GNA11**, and **CAMSAP3**, are associated with cell adhesion, signaling, and structural organization. Reduced expression of these genes may contribute to alterations in vascular and cellular functions that are commonly observed during dengue infection.

The enrichment analyses supported these observations. GO analysis revealed significant enrichment of processes related to humoral immune response, complement activation, blood coagulation, and regulation of vascular permeability. Similarly, KEGG pathway analysis identified enrichment of Complement and coagulation cascades, Cell adhesion molecules, ECM–receptor interaction, Integrin signaling, and Calcium signaling pathways. Together, these findings indicate coordinated changes in immune, inflammatory, coagulation, and vascular-related processes during dengue infection.

Based on both differential expression and functional enrichment results, **IFI27**, **IGHG1**, **IGLV3-21**, **IGLC2**, **JCHAIN**, **MKI67**, **MYBL2**, **LAMA5**, and **JAG2** emerged as the most relevant candidate transcriptomic biomarkers. These genes were selected because they represent the major biological processes identified in this study, including antiviral responses (**IFI27**), humoral immunity (**IGHG1**, **IGLV3-21**, **IGLC2**, and **JCHAIN**), immune cell proliferation (**MKI67** and **MYBL2**), and vascular and cell adhesion functions (**LAMA5** and **JAG2**). Collectively, these genes capture the key molecular changes associated with dengue infection and may serve as promising candidates for future validation studies.

CHAPTER VI

CONCLUSION

This study explored the transcriptomic changes associated with dengue infection to better understand the host response and identify potential biomarkers. Differential gene expression analysis revealed several significantly upregulated and downregulated genes, reflecting changes in immune activity and other important cellular processes. GO and KEGG enrichment analyses further showed that these genes are involved in key biological functions such as immune response, complement activation, coagulation, inflammation, and cell signaling, all of which are known to play important roles in dengue infection.

To make the analysis more accessible, an interactive Shiny application was developed, allowing users to explore gene expression patterns, differential expression results, and functional enrichment through a simple interface with customizable filtering options.

Overall, the findings of this study improve our understanding of the molecular changes that occur during dengue infection and highlight several genes and pathways that may serve as potential biomarkers. Although further validation using larger datasets and experimental studies is needed, this work provides a useful foundation for future research on dengue diagnosis, disease severity, and biomarker discovery.

REFERENCES

1. Dhole P, Zaidi A, Nariya HK, Sinha S, Jinesh S, Srivastava S. Host immune response to dengue virus infection: Friend or foe? *Immuno*. 2024;4:549-577. doi:10.3390/immuno4040033.
2. Krishnamoorthy P, Raj AS, Kumar P, Das N, Kumar H. Host and viral non-coding RNAs in dengue pathogenesis. *Rev Med Virol*. 2022;32(6):e2360. doi:10.1002/rmv.2360.
3. Paz-Bailey G, Adams LE, Deen J, Anderson KB, Katzelnick LC. Dengue. *Lancet*. 2024;403(10427):667-682. doi:10.1016/S0140-6736(23)02592-4.
4. Stark R, Grzelak M, Hadfield J. RNA sequencing: The teenage years. *Nat Rev Genet*. 2019;20:631-656. doi:10.1038/s41576-019-0150-2.
5. Falconi-Agapito F, Kerkhof K, Merino X, Bakokimi D, Torres F, Van Esbroeck M, et al. Peptide biomarkers for the diagnosis of dengue infection. *Front Immunol*. 2022;13:793882. doi:10.3389/fimmu.2022.793882.
6. Thach TQ, Eisa HG, Ben Hmeda A, Faraj H, Thuan TM, Abdelrahman MM, et al. Predictive markers for the early prognosis of dengue severity: A systematic review and meta-analysis. *PLoS Negl Trop Dis*. 2021;15(10):e0009808. doi:10.1371/journal.pntd.0009808.
7. Vasey B, Shankar AH, Herrera BB, Becerra A, Xhaja K, Echenagucia M, et al. Multivariate time-series analysis of biomarkers from a dengue cohort offers new approaches for diagnosis and prognosis. *PLoS Negl Trop Dis*. 2020;14(6):e0008199. doi:10.1371/journal.pntd.0008199.
8. Pradeep SP, Venkatesh PH, Manchala NR, Veedu AV, Basavaraju RK, Selvasundari L, et al. Innate immune cytokine profiling and biomarker identification for outcome in dengue patients. *Front Immunol*. 2021;12:677874. doi:10.3389/fimmu.2021.677874.
9. Moallemi S, Tedla N, Sigera C, Weeratunga P, Fernando D, Rajapakse S, et al. Early circulating biomarkers to predict plasma leakage in dengue fever. *J Infect*. 2025. doi:10.1016/j.jinf.2024.106401.
10. Shukla S, Jadhav SM, Gurav YK, Parashar D, Alagarasu K. Serum ferritin level as a prognostic biomarker for predicting dengue disease severity: A systematic review and meta-analysis. *Rev Med Virol*. 2023;33:e2468. doi:10.1002/rmv.2468.
11. Bernal C, Ping S, Rojas A, Caballero O, Stittleburg V, de Guillén Y, et al. Serum biomarkers and anti-flavivirus antibodies at presentation as indicators of severe dengue. *PLoS Negl Trop Dis*. 2023;17(2):e0010750. doi:10.1371/journal.pntd.0010750.

12. Correa Fierro RE, Mogollón Salazar NG, Cárdenas WB, Medina-Villamizar EJ, Pastuña-Fasso J, Ochoa-Ocampo M, et al. Dengue virus infection in children: Serum lipidomics profiling for biomarker discovery. *PLoS Negl Trop Dis*. 2025;19(11):e0013691. doi:10.1371/journal.pntd.0013691.
13. Suppiah J, Md Sani SS, Hassan SS, Nadzar NIF, Ibrahim NI, Thayan R, et al. Unraveling potential gene biomarkers for dengue infection through RNA sequencing. *Virus Genes*. 2025;61:26-37. doi:10.1007/s11262-024-02114-2.
14. Xie LM, Yin X, Bi J, Luo HM, Cao XJ, Ma YW, et al. Identification of potential biomarkers in dengue via integrated bioinformatic analysis. *PLoS Negl Trop Dis*. 2021;15(8):e0009633. doi:10.1371/journal.pntd.0009633.
15. Krishnamoorthy P, Raj AS, Kumar H. Machine learning-driven blood transcriptome-based discovery of SARS-CoV-2 specific severity biomarkers. *J Med Virol*. 2023;95(2):e28488. doi:10.1002/jmv.28488.

PLAGIARISM REPORT

Identification of Transcriptomic Biomarkers for Dengue Infection

ORIGINALITY REPORT

10 %	6 %	8 %	6 %
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	Submitted to Visvesvaraya Technological University Student Paper	3 %
2	www.bioinforesearch.com Internet Source	1 %
3	Izreen Izzati Razalli, Muhammad-Redha Abdullah-Zawawi, Rabiatal Adawiah Zainal Abidin, Sarahani Harun et al. "Identification and validation of hub genes associated with biotic and abiotic stresses by modular gene co-expression analysis in Oryza sativa L.", Scientific Reports, 2025 Publication	1 %
4	eprints.soton.ac.uk Internet Source	1 %
5	topsecretapiaccess.dovepress.com Internet Source	1 %
6	www.preprints.org Internet Source	1 %
7	ega-archive.org Internet Source	<1 %
8	www.frontiersin.org Internet Source	<1 %
9	Submitted to University of Sussex Student Paper	<1 %

10	Submitted to Rijksuniversiteit Groningen - Tii Student Paper	<1 %
11	www.biorxiv.org Internet Source	<1 %
12	Flávia Heloísa da Silva, Gabriella Lopes Cappellaro, Matheus Felipe Zazula, Mustafa Munir Mustafa Dahleh et al. "Adjuvant-Induced Arthritis Triggers Systemic Inflammatory and Redox Alterations Across Multiple Tissues in Wistar Rats", Springer Science and Business Media LLC, 2026 Publication	<1 %
13	www.mdpi.com Internet Source	<1 %

Exclude quotes On
Exclude bibliography On

Exclude matches < 14 words